

SAP How-To Guide: Extend MDG-M Data Model by a New Entity Type (Reuse Option)



Applies to:

SAP MDG-M running from SAP ECC 6 EhP 6 Master Data Governance. For more information, visit the Master Data Governance homepage. (<http://scn.sap.com/community/mdm/master-data-governance>)

Summary

SAP Master Data Governance provides an out-of-the box solution for the central management of various master data objects such as financial objects, supplier and material. But SAP Master Data Governance also provides the flexibility to customize the solution, in cases where the pre-delivered content does not fully match customer requirements. You can use this guide to extend the MDG-M data model by a new entity. The attribute values of the new entity type will be copied to the corresponding customer tables (reuse option) after activation of the change request by using BADl MDG_BS_MAT_API_SEGMENTS_EXT.

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1. Introduction

SAP Master Data Governance (MDG) is used for embedded MDM, that is, out-of-the-box, domain-specific master data governance to centrally create, change and distribute master data with a focus on SAP Business Suite.

Domain specific content (data models, user interfaces, workflows) is provided as part of the standard for several application areas. It is a common requirement from customers to adapt the MDG data models to their specific needs.

This guide explains how to extend the MDG-M data model by a new entity type by using BAdI MDG_BS_MAT_API_SEGMENTS_EXT. The attribute values of the new entity type will be copied to the corresponding customer tables (reuse option) after activation of the change request.

2. Prerequisites

You must have access to an ERP 6 EhP6 system with the MDG-M business functions enabled. You need access to the system used for cross-client maintenance. You must have basic understanding of the following areas:

- SAP ERP
- SAP MDG
- SAP Floorplan Manager (FPM)

3. Scenario

3.1. High Level Requirements

The business requires the new entity type *Business Partner Details* to save Business Partner Nicknames for materials as part of the MDG-M Material data model.

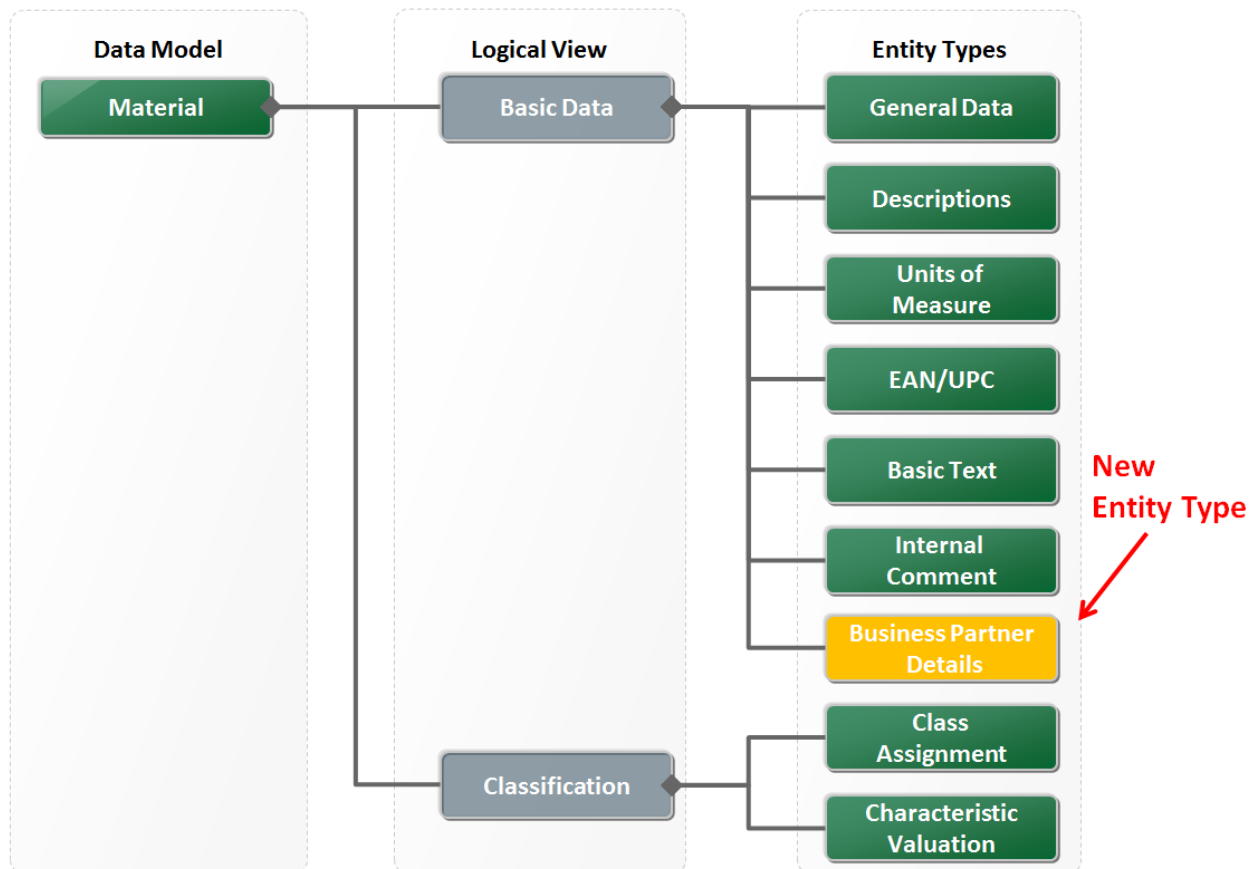


Figure: Data Model – Material (Scope of 2011 Delivery) with custom entity type “Business Partner Details”

3.2. Governance Process

The default governance process delivered with MDG will be used. No changes to the governance process are necessary as part of this scenario.

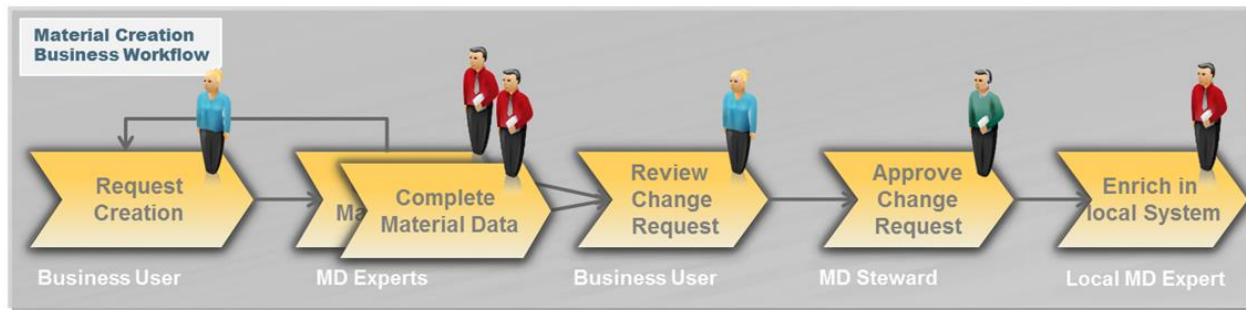


Figure: Material Processing in Master Data Governance

4. Introduction

The implementation steps in this document are easier to understand if you are familiar with the basic concepts; in particular data modeling and user interface customization. In this introduction section, you can learn more about these concepts. Alternatively you can skip this section and move straight to the implementation part.

4.1. Data Modeling

In MDG, the data model is a central part of the application. SAP delivers several preconfigured data models that you can start using with little configuration.

4.1.1. Basic Data Model for Material

Looking at the *Material* object type and its related data in an abstract way, you can distinguish the following categories of data fields:

- Identifying Material Data ~ 10 fields
- Descriptive Material Data ~ 100 fields
- Process Controlling Material Data ~ 1.000 fields

MDG for material data focuses on the main identifying, descriptive attributes and process controlling data of the Material Master.

It is important to understand that MDG not only delivers the data fields in a model, but also comes with the standard business rules to check for completeness and consistency. These checks are only enforced when necessary in the process.

You can either centralize the maintenance of process controlling data on the MDG hub by using the standard backend transactions for material maintenance or you can decentralize the maintenance of process controlling data.

If customers require centralized governance of more process controlling data, they can enhance the delivered model (for example, they can add plant data or sales organization data).

The delivered standard data model is **MM**. (This model is linked to **Material** business object ID: **194**. It is also linked to the **Change Request** business object type: **BUS2550**.)

The **MM** data model for the different releases can be found here:

<http://www.sdn.sap.com/irj/scn/index?rid=/library/uuid/e03cb2f0-3e03-3010-cca5-d4cd11592c28>

4.1.2.1. Storage and Use Types

You assign storage and use type to specify whether and how master data can be changed in Master Data Governance. The storage and use type also indicates which database tables are generated by the system.

Changeable via Change Request; Generated Database Tables (Type 1)	The master data of this storage and use type can be changed in Master Data Governance with a change request. The system generates all necessary database tables: check and text tables as well as additional tables, for example, for attachments and sets. The common key fields of these tables are: • The entity type itself • The edition – if you previously specified in the data model that the validity of master data changes is restricted to editions • The entity types that are assigned to the entity type through leading relationships Furthermore, all tables contain a checkbox that indicates whether the master data record is active. Depending on the workflow template used, it is possible that a master data record is not set to Active until the change request in which the record was created or changed is released. The settings you make for the entity type (such as language dependency) result in additional key fields in the text table and the tables for attachments and sets. The non-key fields contained in the text table are the entity texts. The non-key fields contained in the check table are the attributes of the entity type. The attachment and set tables contain predefined non-key fields. Furthermore, all database tables contain a checkbox that indicates whether the master data record was deleted. The check table also contains attributes that record which user created or changed the data records and when this was done.
Changeable w/o Change Request; Generated Check/Text Tables (Type 2)	The master data of this storage and use type can be changed in Master Data Governance without a change request. The system generates only the check and text tables with the entity type as well as with the entity types assigned to the entity type through leading relationships as fixed key fields. The non-key fields contained in the text table are the entity texts. The check table does not contain non-key fields.
Not Changeable via MDG; No Generated Tables (Type 3)	The master data of this storage and use type cannot be changed in Master Data Governance. Therefore, the system does not generate database tables. Instead, the system derives the available values from the domain that is assigned to the data element – either from the assigned value table or from the domain fixed values.
Changeable via Other Entity Type; Generated Database Tables (Type 4)	The master data of this storage and use type can be changed in Master Data Governance only with a change request of an entity type with storage and use type 1. The entity type needs to be in a relationship with the relationship type leading and assigned as the To-entity type to an entity type with storage and use type 1. The system generates the check table as described for storage and use type 1, but also generates the entity types that are assigned through qualifying relationships as key fields. The system does not generate a text table, attachments, or sets since entity texts are not allowed for entity types with this storage and use type.

You can view the settings for Storage and use Type for existing Entity Types in Customizing for *Master Data Governance* under **General Settings > Data Modeling > Edit Data Model**. You select the **MM** data model and double click on **Entity Types** (view cluster **VC_USMD001**). In the list of entity types you can double click an entity type to view its details as shown below for Entity Type **MATERIAL**.

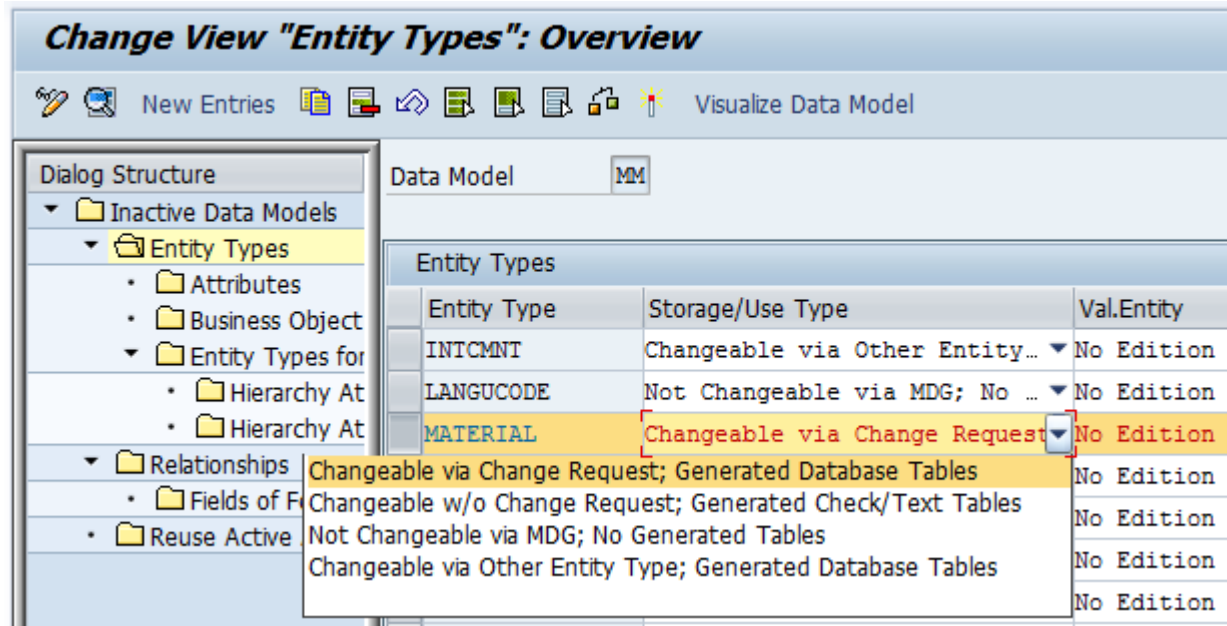


Figure: Selection box showing the different storage/use types in MDG

The table below gives you some criterions for the decision between storage and use type 1 and 4.

Criterion	Type1	Type4
Cross-dependencies	Need to handle dependencies between Type1 entities (include into CR, blocking, ...)	Easy to implement
Parallel CRs	Not needed	Necessary if two Type4 entities for same Type1 entity shall be maintained independently
Snapshot	Independent	Together with corresponding Type1 entity and all other Type4 entities (→ Performance)
Enqueue („Sperschatten“)	This entity plus corresponding Type4 entities	This entity, corresponding parent entity and all its child entities
Authorization	Own authorization objects	Depends on authorization object of Type1 entity (plus maybe additional own ones)
Implementation effort	Own access class	MDG_BS_MAT_API_SEGMENTS_EXT BAdI (lean, no CD mapping, enqueue, Query, ...) or own access class
Mass Maintenance	OK	Not possible (exception: 1:1 Relationship)

Multi-Record Processing	No difference	
Use case	• Production Version (MKAL) • Document Assignment (DRAD)	The rest of the material entities (for example MARC or QMAT)

Note

It's usually not meaningful to define several Type1 entities with the same key (example: MARSALES/-PURCH).

Important

For more complex requirements like search, mapping, change documents, field properties, authorization, locks or derives, you need to provide an own access class by implementing interface `if_usmd_pp_access` instead of using `MDG_BS_MAT_API_SEGMENTS_EXT` BAdI. This guide only describes the usage of the BAdI `MDG_BS_MAT_API_SEGMENTS_EXT`.

4.1.2.2. Relationship Type

If you have defined multiple entity types, you can determine what type of relationship should link them (leading, referencing, qualifying, or foreign key relationship). For each relationship, you specify a relationship type and cardinality.

Relationship Type	Definition
Referencing	Specifies the <i>From-Entity type</i> as an attribute of the <i>To-Entity type</i> .
Leading	Specifies the <i>From-Entity type</i> on a higher level than the <i>To-Entity type</i> . The <i>From-Entity type</i> is automatically taken as the key in the generated tables. A <i>Leading</i> relationship type is identical to a <i>Qualifying</i> relationship type, except when the <i>To-Entity type</i> has a <i>Storage and Use Type</i> of 4. Master data for <i>To-Entity types</i> in <i>Leading</i> relationships is processed in the context of the entity type that is assigned using the leading relationship.
Qualifying	Specifies the <i>From-Entity type</i> on a higher level than the <i>To-Entity type</i> . The <i>From-Entity type</i> is automatically taken as the key in the generated tables.

The following options are possible for the relationship between two entity types:

Cardinality	Definition
1:N	This cardinality represents a mandatory relationship in which one or more <i>To-Entity Types</i> can be assigned to a <i>From-Entity Type</i> . This cardinality is valid for relationships with the relationship types <i>Leading</i> , <i>Qualifying</i> , and <i>Referencing</i> . In relationships with the relationship type <i>referencing</i> , the <i>From-Entity Type</i> is a required attribute of the <i>To-Entity Type</i> .
0:N	This cardinality represents an optional relationship in which any number <i>To-Entity Types</i> can be assigned to a <i>From-Entity Type</i> .

Note

Which relationship types are permitted depends on the storage and use types of the entity types. For a table with detailed information refer to help.sap.com.

Important

The general design assumption is that there is a 1:N relationship between a database table and its entity types. This means one entity type does not bundle several database tables.

4.1.2.3. Reuse Area versus the Flexible Option

When you extend the SAP delivered data model by a new entity type you must decide where to store data after the activation of a change request. During the processing of the change request the system stores data in the MDG staging area. After the activation of a change request, you can move the data to tables outside of MDG or keep the data in the MDG tables.

For optimal integration into SAP Business Suite, MDG provides the following two persistence modes:

- Generated active area (flex mode) – Tables as defined in the MDG data model are used to store active data.
- Reuse active area (re-use mode) – Existing structures of applications are used. For example, MDG for material makes use of the **MARA** table in ECC.

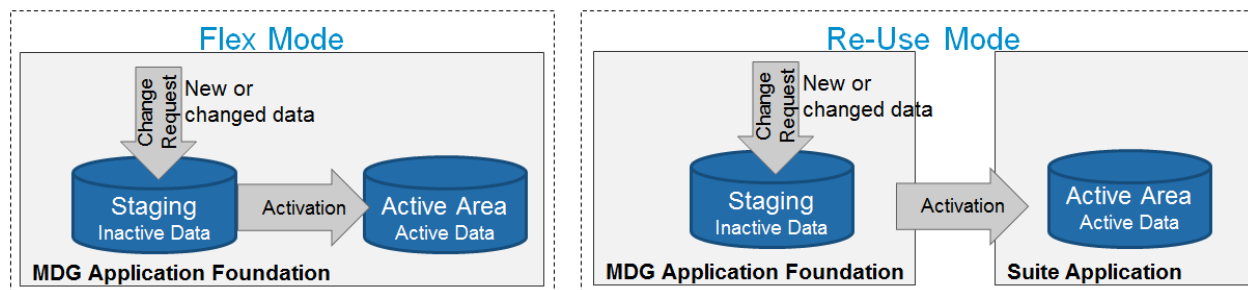


Figure: Flexibility Option (left) versus Reuse Option (right)

Where the data is stored is specified by the **Reuse Area** setting on the **Data Model** or **Entity Type** level as shown in the screenshots below.

Change View "Inactive Data Models": Overview

New Entries Visualize Data Model

Inactive Data Models		
Data Model	Description (medium text)	Reuse Area
0F	Chart of Accounts	
0G	Chart of Accounts and Org. Units	
BP	Business Partner	PARTNER
MM	Material Maintenance MM01/02	MATERIAL
TD	Flight Data Model f. Field Property Test	TD
Z1	Copy of MM for message processing	MATERIAL
Z2	Copy of MM for message processing	MATERIAL
Z9	test	

Dialog Structure:

- Inactive Data Models
 - Entity Types
 - Attributes
 - Business Object
 - Entity Types for
 - Hierarchy At
 - Hierarchy At
 - Relationships
 - Fields of Foreign
 - Reuse Active Areas

Figure: Assignment of Reuse Area for the Data Model **MM**

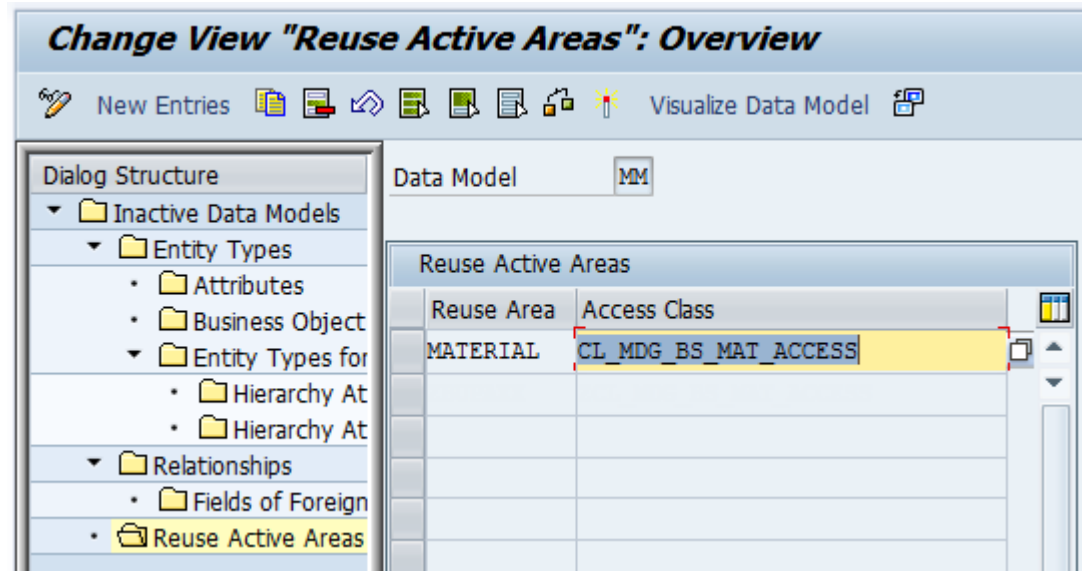


Figure: Assignment of Access Class for Reuse Area **MATERIAL**

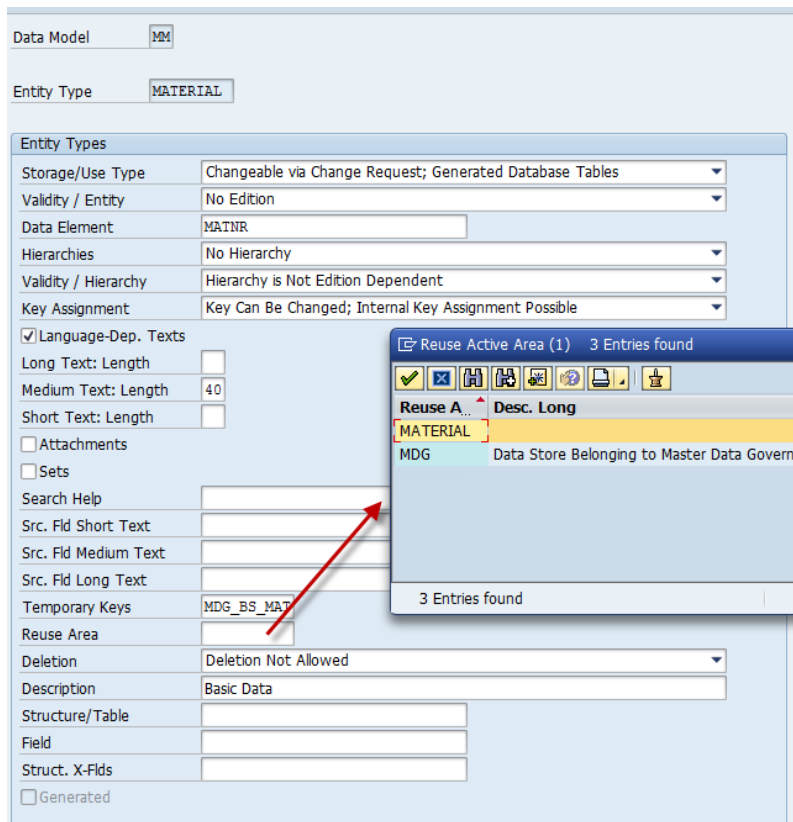


Figure: Alternative assignment of Reuse Area on Entity Type level

The MDG model **MM** is preconfigured with one reuse area called **MATERIAL**. This reuse area points to the access class **CL_MDG_BS_MAT_ACCESS**, which can handle all fields of the pre-delivered data model and some more.

If you extend the data model by a new entity type and want the data of that entity type to remain in the MDG tables after activation you can choose MDG as a reuse area.

The table below gives you some criterions for the decision between flex or reuse mode.

Criterion	Flex	Reuse (Z table)
Direct Update of Active Area (for migration, ...)	Not possible	Possible and used in IDoc/DIF/LSMW/API/...
DIF usage	Not recommended for MDG-M (can't write to active area)	OK
SOA	Not recommended for MDG-M (can't write to active area)	OK
File Upload	No difference (updates only staging)	
Access/usage in productive applications (in co-deployment) like • Search • Typed read access • ...	Dynamic tables, access only via IF_USMD_MODEL_EXT-> READ_CHAR_VALUE (or complex select), no indices on DB level	Can be tuned via indices, view definition possible and transportable, ...
Mass Maintenance/Multi-Record Processing	No difference	
Performance in maintenance	• Flex entities need no snapshot (and snapshot comparison) • Number of records in active area has impact on performance	• Reuse entities need snapshot • Number of records in reuse active area has no impact on performance
DRF usage	No difference	
Project effort	No additional coding required (plus optional additional checks, feeder class, ...)	Coding needed – either implementation of BADI MDG_BS_MAT_API_SEGM ENTS_EXT or specific access class
Use case	Co-Deployment: Stays in MDG, no need to access data operatively	Co-Deployment: Needs to be accessed by operative processes. Both deployments: Take existing data/extensions under governance Both deployments: Data import must be done without CR for performance reasons (or no governance for migration/import needed)

4.1.2.4. Input Help (Accessible Using the F4 Key)

The system applies the following rules of precedence when assigning input help:

1. Search help assignment in data model definition
2. Backend structure MDG_BS_MAT_S_MARA, MDG_BS_MAT_S_* (not existing for Flex Entities)
 - a. Search help assignment in the structure
 - b. Value table on domain with foreign key association
3. Search help assignment on data element (for flex entities)
4. Fixed values or value table on domain

Note: Value table on domain without foreign key assignment (for flex entities) is not supported out of the box

4.1.2.5. Code Lists

The considered code list for the check comes from the **Fixed Values** or **Value Range** table which is assigned to the domain of the data element.

4.1.2.6. Data Modeling Considerations for List-UIBBs

If you want the Material UI to use two or more list UIBBs (User Interface Building Blocks), you must adjust the data model. You can implement independent list UIBBs or interdependent list UIBBs. For more information, see the table below.

	Desired System Behavior	Action
Independent List UIBBs	Changes to one list UIBB have no impact on the other list UIBB.	Create separate entity types and assign one to each List-UIBB.
Interdependent List UIBBs	If you create a new row for the same key in one UIBB, the system creates a new row in the other UIBB.	Either assign the same entity type for both List-UIBBs or implement a derivation.

Example Implementation of Independent and Interdependent UIBBs

You have created **interdependent** list UIBBs for the following user interfaces using the **ZZMARC** entity type:

- **MRP1/MRP2**
- **Foreign Trade Export**

In the **Component Configuration** for **MRP1/MRP2** and for **Foreign Trade Export**, you have maintained the **ZZMARC** entity type.

In addition, you have created an **independent** list UIBB for **Foreign Trade Export (separate)** user interface, using a separate entity type **ZZMARCFTE**. The reason for this decision is that a particular customer wants only to enter data relevant to foreign trade export, and does not want to enter MRP data. In the **Component Configuration** for **Foreign Trade Export (separate)**, you have maintained the **ZZMARCFTE** entity type. The **ZZMARCFTE** entity type is not affected when you enter a new row in **ZZMARC** component configurations.

The screenshot displays the SAP MDG-M Data Model configuration interface. It is divided into three main sections:

- MRP 1:** Contains a table with columns: Plant, MRP Type, MRP description, MRP Controller, Lot size, and Tot. repl. lead time. The first row has the value '1000' in the Plant column.
- Foreign Trade Export:** Contains a table with columns: Plant, Comm./imp. code no., Description, Country of origin, and Name. The first row has the value '1000' in the Plant column.
- Foreign Trade Export (separate):** Contains a table with columns: Plant, Comm./imp. code no., Description, Country of origin, and Name. The first row has the value '1100' in the Plant column.

Large red watermarks 'ZZMARC' and 'ZZMARCFTE' are overlaid on the MRP 1 and Foreign Trade Export (separate) sections, respectively.

4.2. User Interface Configuration

The User Interface in MDG is configured using the Floorplan Manager. The Floorplan Manager (FPM) is a Web Dynpro ABAP application that provides a framework for developing new Web Dynpro ABAP application interfaces consistent with SAP UI guidelines.

The entry point you need for starting an application is the *application configuration*, which is tied to a single Web Dynpro application. The necessary information needed to start the application is divided between the following two entities:

- **Web Dynpro ABAP Application:** Contains the information about the main component and window of the application
- **Web Dynpro ABAP Application Configuration:** Contains the information about the configuration used for starting the main component

There are only 3 different main components used in FPM-based applications. Each one corresponds to one of the supported floorplans:

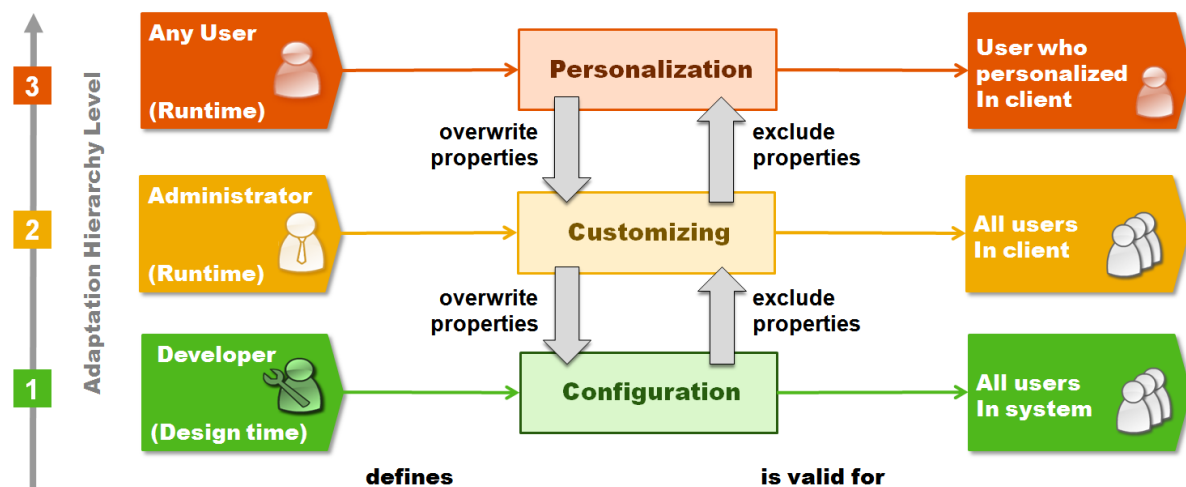
- OIF (Object Instance Floorplan): component **FPM_OIF_COMPONENT**
- GAF (Guided Activity Floorplan): component **FPM_GAF_COMPONENT**
- OVP (Overview Page Floorplan): component **FPM_OVP_COMPONENT**

In the screenshot below application **MDG_BS_MAT_OVP** is using component **FPM_ADAPTABLE_OVP** with configuration **BS_MAT_OVP_CBA** to provide the context-based adaptation (CBA) functionality. The layout is provided by **FPM_OVP_COMPONENT** in combination with configuration **BS_MAT_OVP_LAYOUT**.

Application Configuration BS_MAT_OVP				
Save Cancel Edit Check New Window Enhance Properties Test				
▼ Assign Web Dynpro Component				
Assign Configuration Name				
Component Usage	Component	Implementation	Configuration Name	
▼ MDG_BS_MAT_OVP	FPM_ADAPTABLE_OVP	FPM_ADAPTABLE_OVP	BS_MAT_OVP_CBA	
▪ OVP	FPM_OVP_COMPONENT	FPM_OVP_COMPONENT	BS_MAT_OVP_LAYOUT	

4.2.1. Adaptation Options in Floorplan Manager

A Floorplan Manager UI can be adapted using different techniques. The figure below shows the relationship between configuration, customizing, and personalization. Context-Based-Adaptation is another way the user interface can be customized for specific use cases.



In the context of MDG, you typically choose to *customize* the SAP delivered configuration. Only if customizing is not feasible do you copy the SAP delivered UI configuration to the customer namespace and change the copy.

In the following cases the UI should be copied rather than customized:

- Code changes are required
- The UI needs to be changed for all users in the system and not only client-specific
- The changes to the UI are extensive

Note

For more details regarding options for Floorplan manager user interface adaptation, advantages, disadvantages and steps required please familiarize yourself with SAP Note 1619534.

See also How To Guide for the UI: <http://scn.sap.com/docs/DOC-30192>

4.2.2. Handling Previous Enhancements Using the “UI BAdI”

With EhP6, the Business Add-In previously available to adjust the user interface for single processing of material is no longer available. Instead, a new UI provides greater flexibility while retaining and refining the functionality provided with EhP5. Below is the full list of the functions previously available through the single processing UI by implementing the BAdI in EhP5, that are now provided under the EhP6 UI paradigm:

- Initialize the displayed data (when creating a new entity, for example)
- Restrict the values displayed in a dropdown list field or selection field group
- Restrict the values displayed in the input help
- Dynamically control the visibility of fields on the user interface and of the property that determines if fields are required or display-only
- Define navigation destinations of UI elements of the type hyperlink (or pushbutton)
- Check if the lead selection of a table may be changed

For more information, see the MDG consulting SAP Note 1606341.

4.2.3.Removing Customizing or Personalization

If required, a system administrator can delete customizing or personalization from a central place using the following Web-Dynpro applications, which must be used with caution:

- **WD_ANALYZE_CONFIG_USER**
- **WD_ANALYZE_CONFIG_COMP**
- **WD_ANALYZE_CONFIG_APPL**

5. Implementation

Two major building blocks make up the implementation of the entity type extension. In the first phase, you extend the data model. In the second phase, you extend the user interface to include the new entity type.

The flow diagram below shows the detailed implementation steps. We recommend you use it as an orientation. Each box in the diagram below corresponds to a section in this guide in which you find detailed execution instructions.

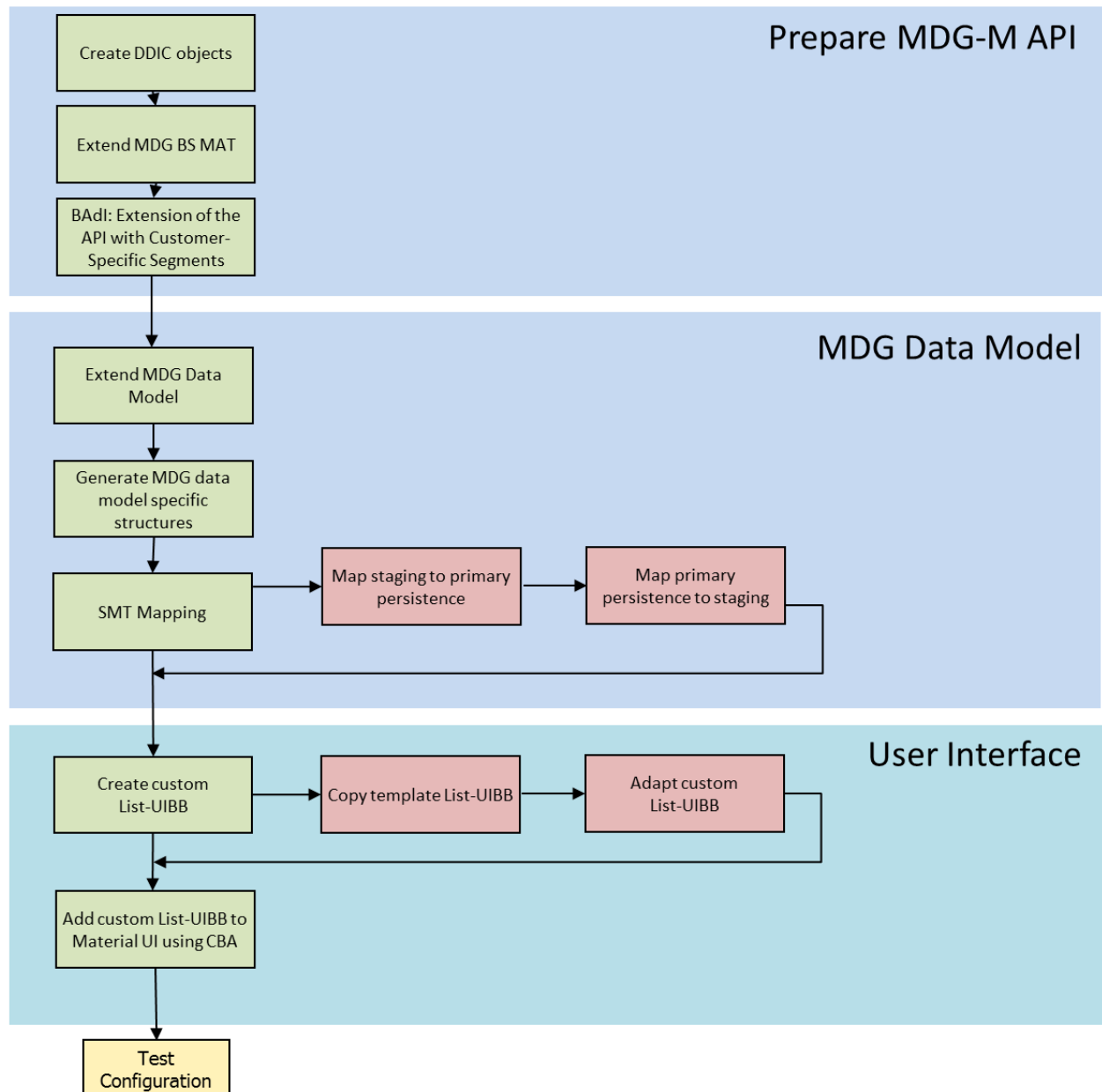


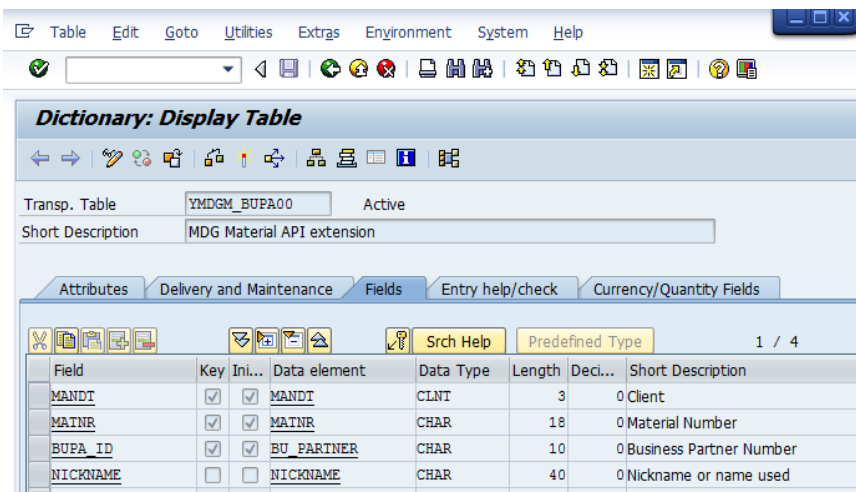
Figure: Implementation steps for re-use Entity-Type extension

5.1. Prepare MDG-M API

5.1.1. Create DDIC Objects

In this section you create the table and structure that you use to store master data after activation in MDG. After activation data will be transferred from the MDG staging area to the table that you define in the following steps.

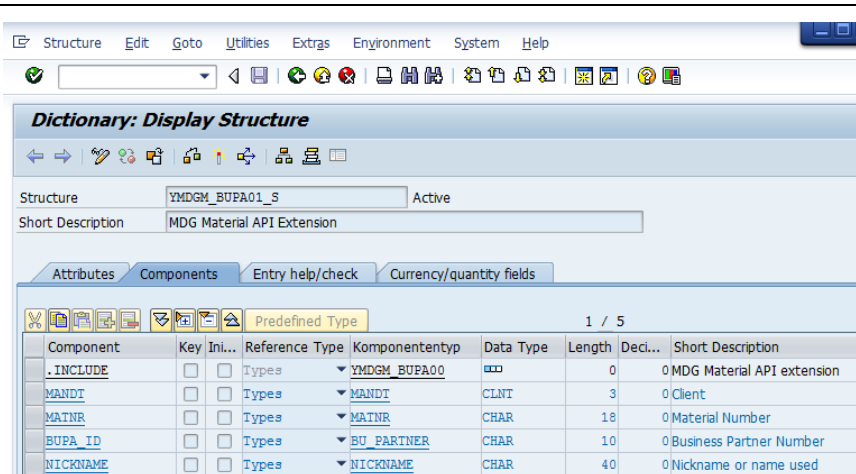
1. In transaction **SE11** create table **YMDGM_BUPA00** as shown.



The screenshot shows the SAP SE11 Dictionary: Display Table for table YMDGM_BUPA00. The table is active and has the short description 'MDG Material API extension'. The Fields tab is selected, showing a list of fields with their keys, initial values, data elements, data types, lengths, and decimal places.

Field	Key	Ini...	Data element	Data Type	Length	Deci...	Short Description
MANDT	☒	☒	MANDT	CLNT	3	0	Client
MAINR	☒	☒	MAINR	CHAR	18	0	Material Number
BUPA_ID	☒	☒	BU_PARTNER	CHAR	10	0	Business Partner Number
NICKNAME	☐	☐	NICKNAME	CHAR	40	0	Nickname or name used

2. In transaction **SE11** create structure **YMDGM_BUPA01_S** as shown.



The screenshot shows the SAP SE11 Dictionary: Display Structure for structure YMDGM_BUPA01_S. The structure is active and has the short description 'MDG Material API Extension'. The Components tab is selected, showing a list of components with their keys, initial values, reference types, component types, data types, lengths, and decimal places.

Component	Key	Ini...	Reference Type	Komponententyp	Data Type	Length	Deci...	Short Description
.INCLUDE	☐	☐	Types	YMDGM_BUPA00		0		MDG Material API extension
MANDT	☐	☐	Types	MANDT	CLNT	3	0	Client
MAINR	☐	☐	Types	MAINR	CHAR	18	0	Material Number
BUPA_ID	☐	☐	Types	BU_PARTNER	CHAR	10	0	Business Partner Number
NICKNAME	☐	☐	Types	NICKNAME	CHAR	40	0	Nickname or name used

3. In transaction **SE11** create table type **YMDGM_BUPA01_T** as shown.

Table Type: YMDGM_BUPA01_T Active

Short text: MDG Material API extension

Attributes | Line Type | Initialization and Access | Key | Sekundärschlüssel

☒ Line Type: YMDGM_BUPA01_S

☐ Predefined Type

Data Type: []

No. of Characters: 0 Decimal Places: 0

☐ Reference type

☐ Name of Ref. Type: []

☐ Reference to Predefined Type

Data Type: []

Length: 0 Decimal Places: 0

4. Make sure table type **YMDGM_BUPA01_T** is a **Sorted Table**.

Table Type: YMDGM_BUPA01_T Active

Short text: MDG Material API extension

Attributes | Line Type | Initialization and Access | Key

Initial Line Number: 0

Access

☒ Sorted Table

☐ Standard Table

☐ Hashed Table

☐ Index Table

☐ Not Specified

5. Make sure table type **YMDGM_BUPA01_T** uses the keys as shown.

Dictionary: Display Table Type

Table Type: **YMDGM_BUPA01_T** Active

Short text: Table type for for table YMDGM_BUPA00

Attributes | Line Type | Initialization and Access | **Primary Key** | Second

Key definition

☐ Standard Key
☐ Line Type
☒ Key Components
☐ Not specified

Key category

☒ Unique
☐ Non-Unique
☐ Not specified

Choose components

Key component

MATNR
BUPA_ID

6. In transaction **SE11** create structure **YMDGM_BUPA01_S_X** as shown.

Note:
DELETE_ROW
 indicates that this
 row shall be deleted

Dictionary: Display Structure

Structure: **YMDGM_BUPA01_S_X** Active

Short Description: MDG Material API Extension test(BUPAXX) Table X structure

Attributes | **Components** | Entry help/check | Currency/quantity fields

Predefined Type

Component	Reference Type	Komponententyp	Data Type	Length	Deci...
<u>M</u> ATNR	1 Types	▼ <u>M</u> ATNR	CHAR	18	
<u>B</u> UPA_ID	1 Types	▼ <u>B</u> U_PARTNER	CHAR	10	
<u>N</u> ICKNAME	1 Types	▼ <u>X</u> FELD	CHAR	1	
<u>D</u> ELETE_ROW	1 Types	▼ <u>X</u> FELD	CHAR	1	

7. In transaction **SE11** create table type **YMDGM_BUPA01_T_X** as shown.

Dictionary: Display Table Type

Table Type: YMDGM_BUPA01_T_X New

Short text: MDG Material API extension

Attributes Line Type Initialization and Access Key Sekundärschlüssel

☒ Line Type YMDGM_BUPA01_S_X

☐ Predefined Type

Data Type: []

No. of Characters: 0 Decimal Places: 0

☐ Reference type

☐ Name of Ref. Type: []

☐ Reference to Predefined Type

Data Type: []

Length: 0 Decimal Places: 0

8. Make sure table type **YMDGM_BUPA01_T_X** is a sorted table.

Dictionary: Display Table Type

Table Type: YMDGM_BUPA01_T_X Active

Short text: MDG Material API extension

Attributes Line Type Initialization and Access Key Sekundärschlüssel

Initial Line Number: 0

Access

☐ Standard Table

☒ Sorted Table

☐ Hashed Table

☐ Index Table

☐ Not Specified

9. Make sure table type **YMDGM_BUPA01_T_X** has the key fields as shown.

Dictionary: Display Table Type

Table Type: YMDGM_BUPA01_T_X Active
Short text: MDG Material API extension

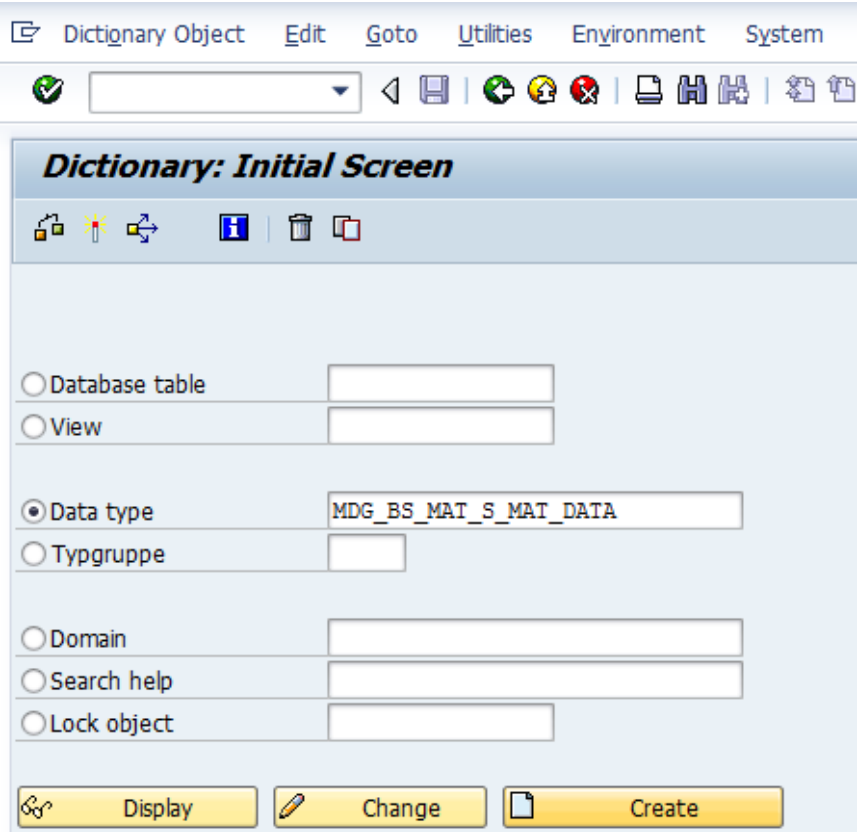
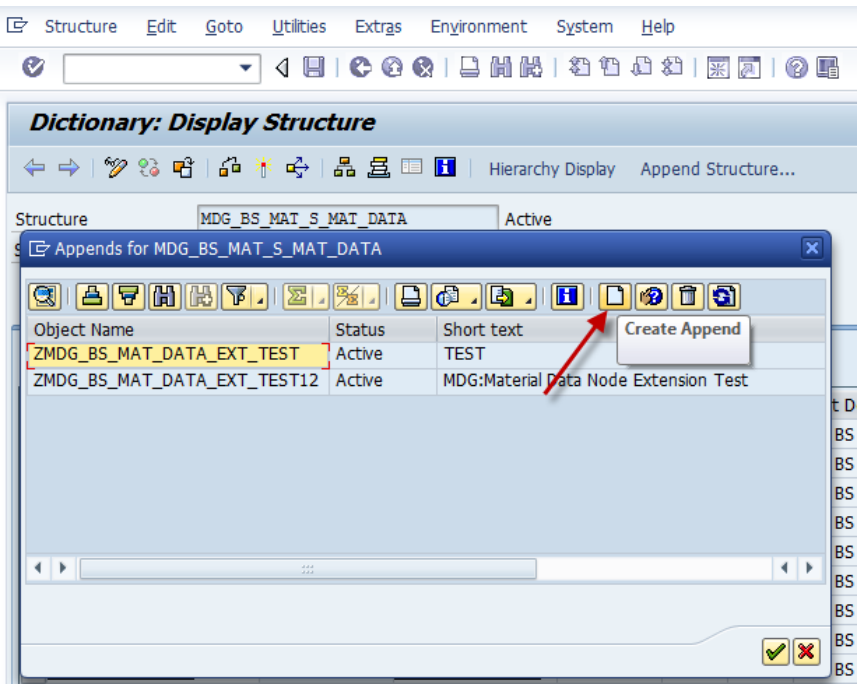
Attributes Line Type Initialization and Access **Key** Sekundärschlüssel

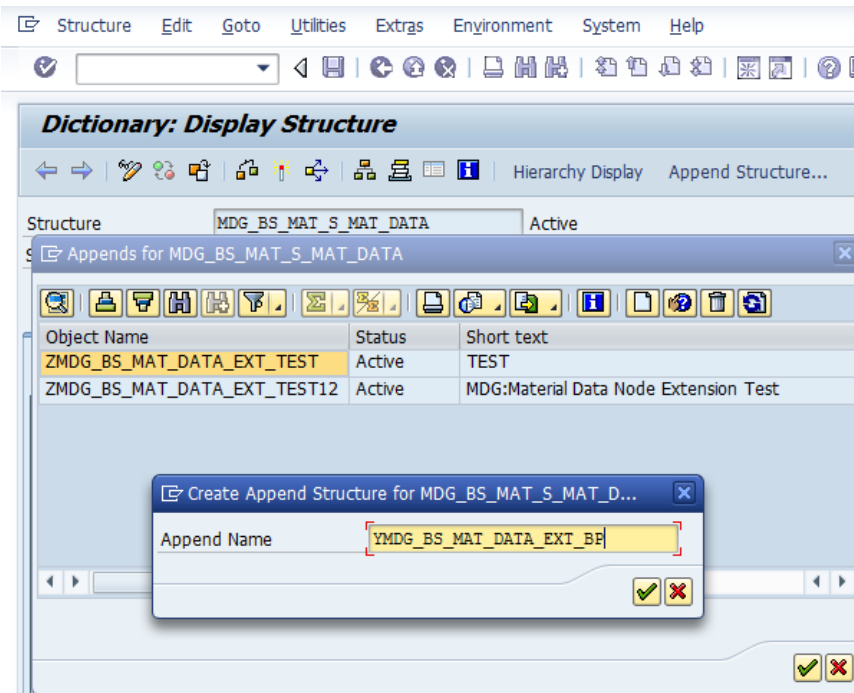
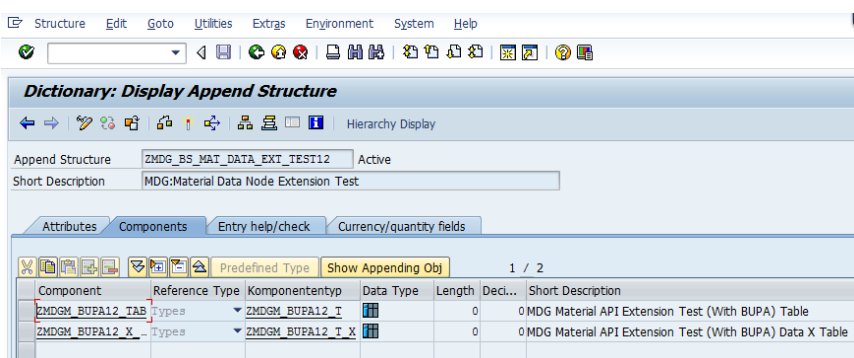
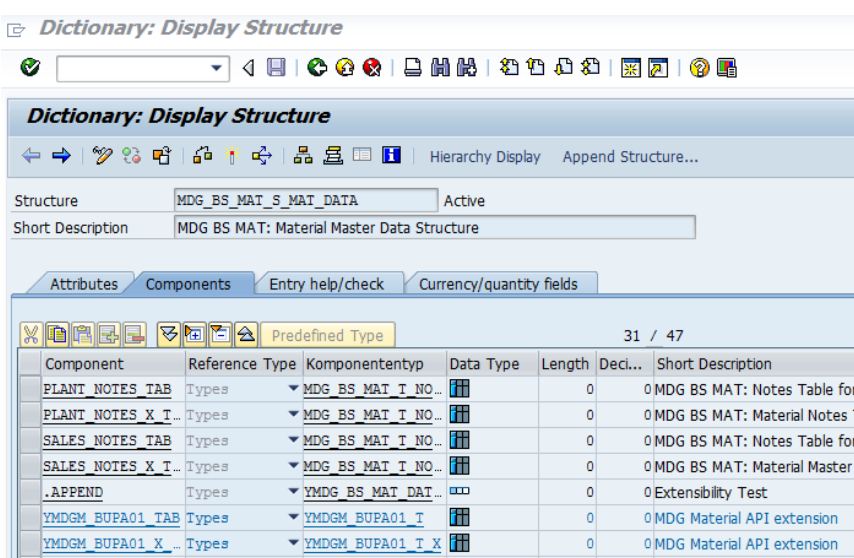
Key definition	Key category
☐ Standard key	☒ Unique
☐ Line type	☐ Non-unique
☒ Key components	☐ Not specified
☐ Not specified	

Choose components

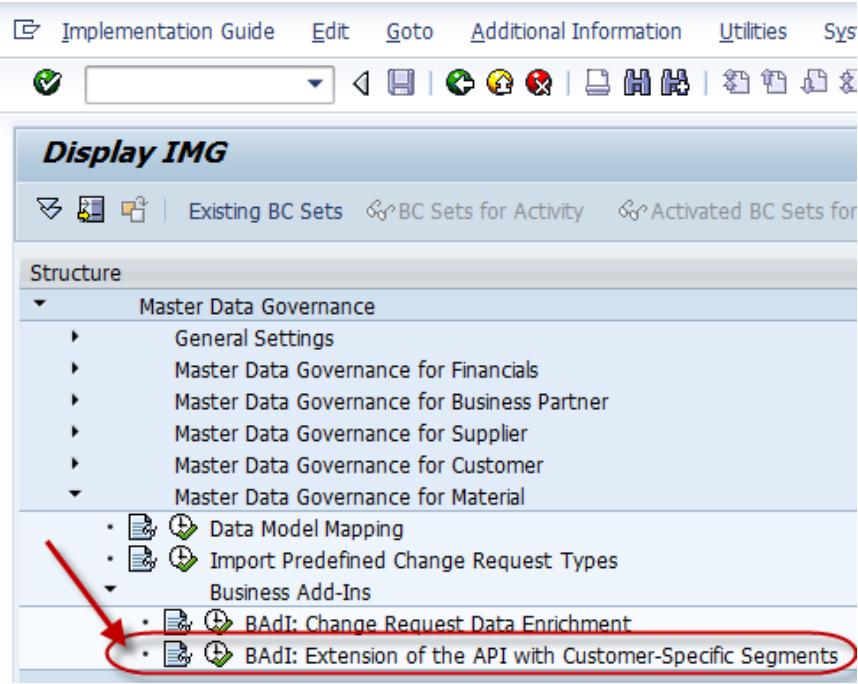
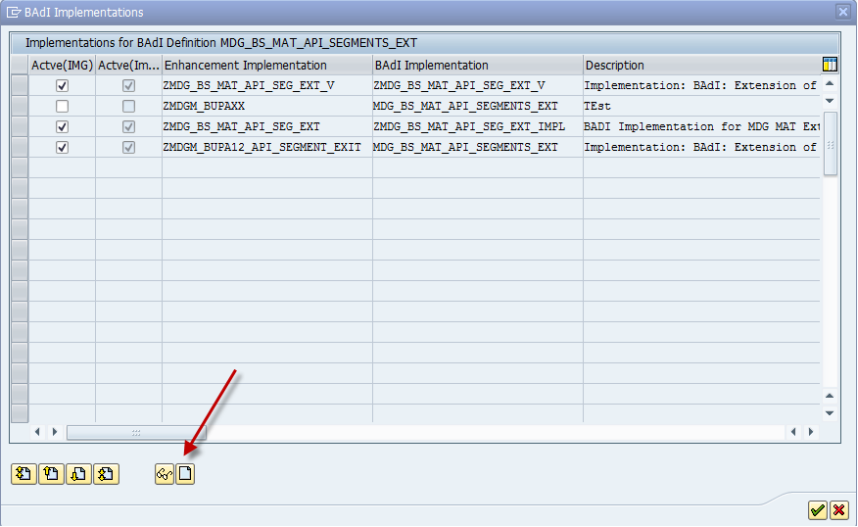
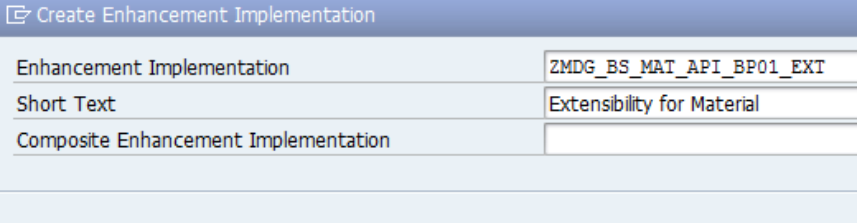
Key component
MATNR
BUPA_ID

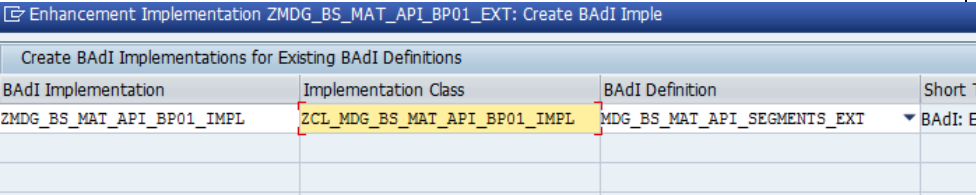
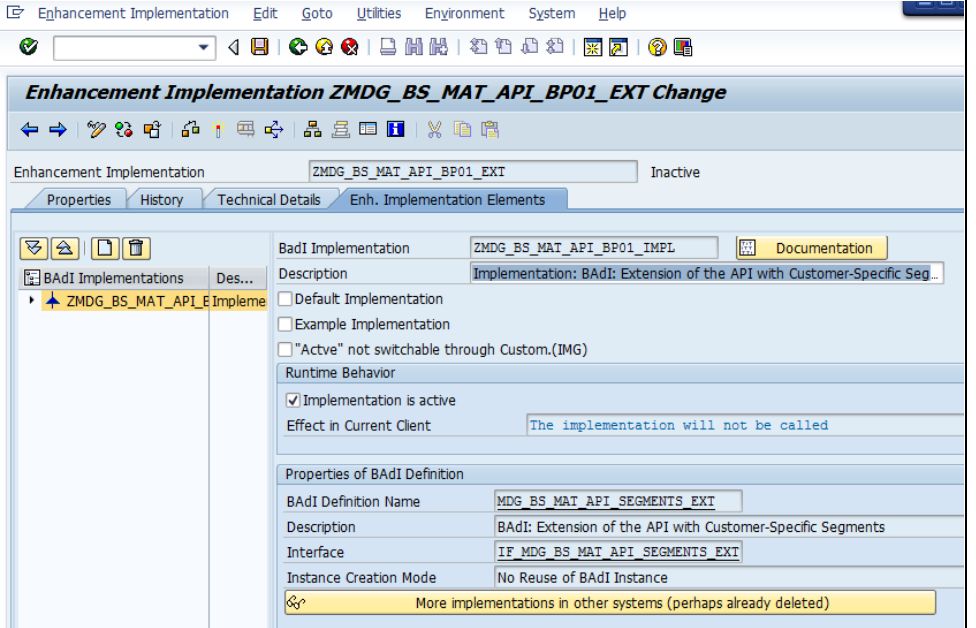
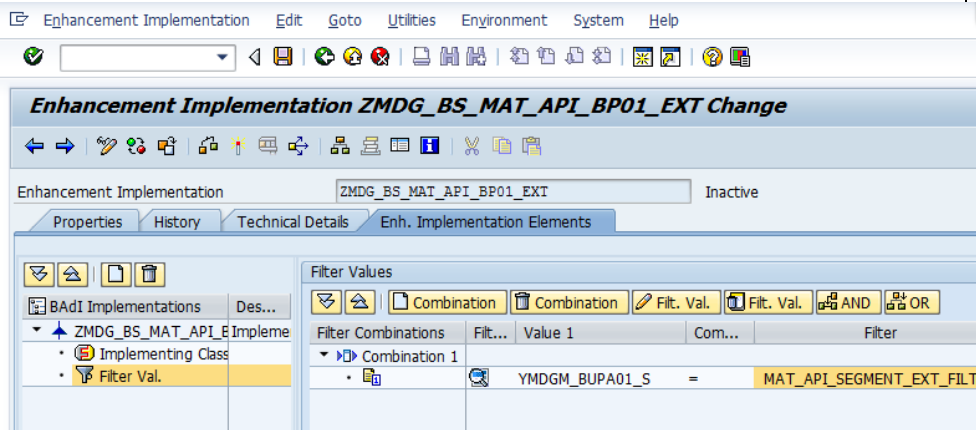
5.1.2. Extend MDG BS MAT (Material Master Data Structure)

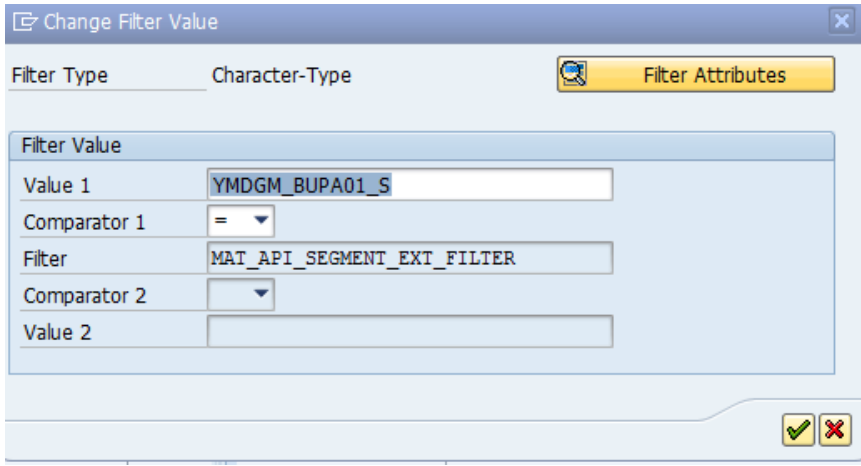
1.	Start transaction SE11. Enter MDG_BS_MAT_S_MAT_DATA as shown. Choose Display.										
2.	Create a new Append.	 <table border="1"> <thead> <tr> <th>Object Name</th> <th>Status</th> <th>Short text</th> </tr> </thead> <tbody> <tr> <td>ZMDG_BS_MAT_DATA_EXT_TEST</td> <td>Active</td> <td>TEST</td> </tr> <tr> <td>ZMDG_BS_MAT_DATA_EXT_TEST12</td> <td>Active</td> <td>MDG:Material Data Node Extension Test</td> </tr> </tbody> </table>	Object Name	Status	Short text	ZMDG_BS_MAT_DATA_EXT_TEST	Active	TEST	ZMDG_BS_MAT_DATA_EXT_TEST12	Active	MDG:Material Data Node Extension Test
Object Name	Status	Short text									
ZMDG_BS_MAT_DATA_EXT_TEST	Active	TEST									
ZMDG_BS_MAT_DATA_EXT_TEST12	Active	MDG:Material Data Node Extension Test									

3.	Enter the Append Name as shown.																																																									
4.	Enter the append component details as shown.	 <table><thead><tr><th>Component</th><th>Reference Type</th><th>Komponententyp</th><th>Data Type</th><th>Length</th><th>Deci...</th><th>Short Description</th></tr></thead><tbody><tr><td>ZMDGM_BUPA12_TAB</td><td>Types</td><td>ZMDGM_BUPA12_T</td><td></td><td>0</td><td></td><td>0 MDG Material API Extension Test (With BUPA) Table</td></tr><tr><td>ZMDGM_BUPA12_X_</td><td>Types</td><td>ZMDGM_BUPA12_T_X</td><td></td><td>0</td><td></td><td>0 MDG Material API Extension Test (With BUPA) Data X Table</td></tr></tbody></table>	Component	Reference Type	Komponententyp	Data Type	Length	Deci...	Short Description	ZMDGM_BUPA12_TAB	Types	ZMDGM_BUPA12_T		0		0 MDG Material API Extension Test (With BUPA) Table	ZMDGM_BUPA12_X_	Types	ZMDGM_BUPA12_T_X		0		0 MDG Material API Extension Test (With BUPA) Data X Table																																			
Component	Reference Type	Komponententyp	Data Type	Length	Deci...	Short Description																																																				
ZMDGM_BUPA12_TAB	Types	ZMDGM_BUPA12_T		0		0 MDG Material API Extension Test (With BUPA) Table																																																				
ZMDGM_BUPA12_X_	Types	ZMDGM_BUPA12_T_X		0		0 MDG Material API Extension Test (With BUPA) Data X Table																																																				
5.	Save your changes and make sure that the append appears in structure MDG_BS_MAT_S_MAT_DATA as shown. Important naming convention: <Tablename>_TAB for the data part and <Tablename>_X_TAB for the change structure with <Tablename> being the name of the database table you want to put under governance.	 <table><thead><tr><th>Component</th><th>Reference Type</th><th>Komponententyp</th><th>Data Type</th><th>Length</th><th>Deci...</th><th>Short Description</th></tr></thead><tbody><tr><td>PLANT NOTES_TAB</td><td>Types</td><td>MDG_BS_MAT_I_NO...</td><td></td><td>0</td><td></td><td>0 MDG BS MAT: Notes Table for</td></tr><tr><td>PLANT NOTES_X_I_</td><td>Types</td><td>MDG_BS_MAT_I_NO...</td><td></td><td>0</td><td></td><td>0 MDG BS MAT: Material Notes 1</td></tr><tr><td>SALES NOTES_TAB</td><td>Types</td><td>MDG_BS_MAT_I_NO...</td><td></td><td>0</td><td></td><td>0 MDG BS MAT: Notes Table for</td></tr><tr><td>SALES NOTES_X_I_</td><td>Types</td><td>MDG_BS_MAT_I_NO...</td><td></td><td>0</td><td></td><td>0 MDG BS MAT: Material Master</td></tr><tr><td>.APPEND</td><td>Types</td><td>YMDG_BS_MAT_DAI...</td><td></td><td>0</td><td></td><td>0 Extensibility Test</td></tr><tr><td>YMDGM_BUPA01_TAB</td><td>Types</td><td>YMDGM_BUPA01_T</td><td></td><td>0</td><td></td><td>0 MDG Material API extension</td></tr><tr><td>YMDGM_BUPA01_X_</td><td>Types</td><td>YMDGM_BUPA01_T_X</td><td></td><td>0</td><td></td><td>0 MDG Material API extension</td></tr></tbody></table>	Component	Reference Type	Komponententyp	Data Type	Length	Deci...	Short Description	PLANT NOTES_TAB	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Notes Table for	PLANT NOTES_X_I_	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Material Notes 1	SALES NOTES_TAB	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Notes Table for	SALES NOTES_X_I_	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Material Master	.APPEND	Types	YMDG_BS_MAT_DAI...		0		0 Extensibility Test	YMDGM_BUPA01_TAB	Types	YMDGM_BUPA01_T		0		0 MDG Material API extension	YMDGM_BUPA01_X_	Types	YMDGM_BUPA01_T_X		0		0 MDG Material API extension
Component	Reference Type	Komponententyp	Data Type	Length	Deci...	Short Description																																																				
PLANT NOTES_TAB	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Notes Table for																																																				
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SALES NOTES_TAB	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Notes Table for																																																				
SALES NOTES_X_I_	Types	MDG_BS_MAT_I_NO...		0		0 MDG BS MAT: Material Master																																																				
.APPEND	Types	YMDG_BS_MAT_DAI...		0		0 Extensibility Test																																																				
YMDGM_BUPA01_TAB	Types	YMDGM_BUPA01_T		0		0 MDG Material API extension																																																				
YMDGM_BUPA01_X_	Types	YMDGM_BUPA01_T_X		0		0 MDG Material API extension																																																				

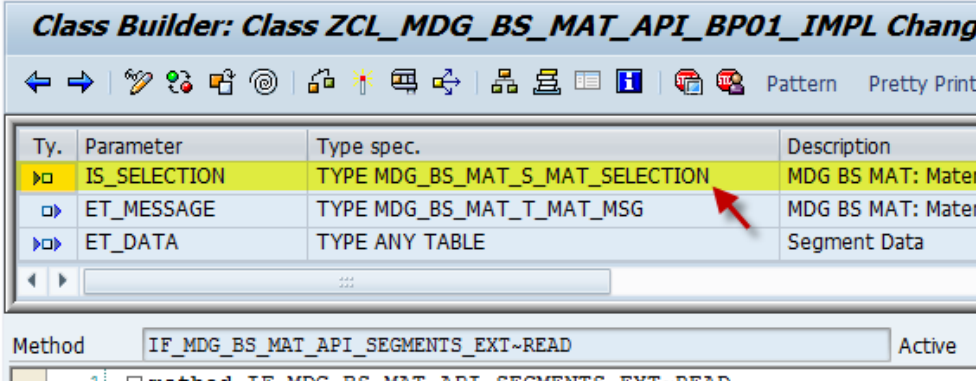
5.1.3.BAdI: Extension of the API with Customer-Specific Segments

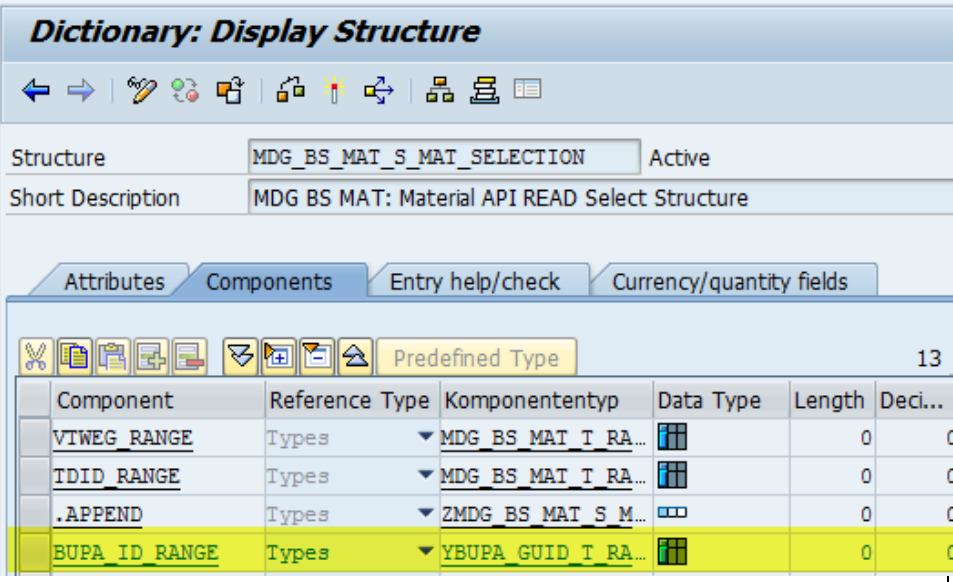
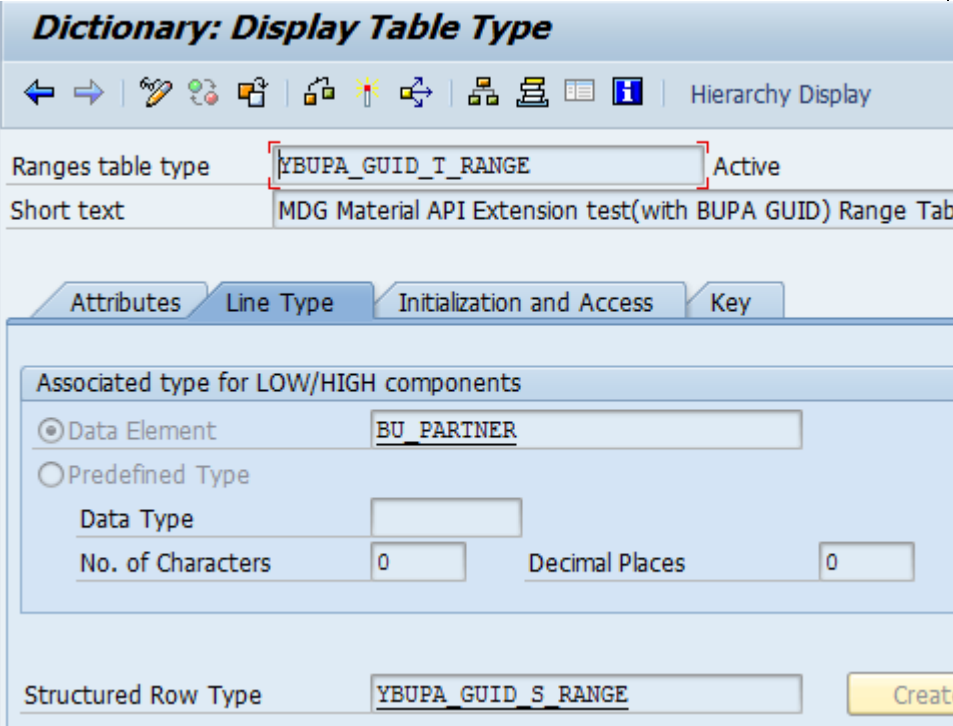
1.	In MDG customizing start the selected entry as shown.	
2.	Create a new BAdI implementation.	
3.	Enter the name of the Enhancement Implementation as shown.	

4.	Enter further details for the BAdI as shown in the screenshot.	 Enhancement Implementation ZMDG_BS_MAT_API_BP01_EXT: Create BAdI Imple Create BAdI Implementations for Existing BAdI Definitions <table><thead><tr><th>BAdI Implementation</th><th>Implementation Class</th><th>BAdI Definition</th><th>Short T</th></tr></thead><tbody><tr><td>ZMDG_BS_MAT_API_BP01_IMPL</td><td>ZCL_MDG_BS_MAT_API_BP01_IMPL</td><td>MDG_BS_MAT_API_SEGMENTS_EXT</td><td>▼ BAdI: E</td></tr></tbody></table>	BAdI Implementation	Implementation Class	BAdI Definition	Short T	ZMDG_BS_MAT_API_BP01_IMPL	ZCL_MDG_BS_MAT_API_BP01_IMPL	MDG_BS_MAT_API_SEGMENTS_EXT	▼ BAdI: E		
BAdI Implementation	Implementation Class	BAdI Definition	Short T									
ZMDG_BS_MAT_API_BP01_IMPL	ZCL_MDG_BS_MAT_API_BP01_IMPL	MDG_BS_MAT_API_SEGMENTS_EXT	▼ BAdI: E									
5.	Check your Enhancement Implementation details look similar to the screenshot.	 Enhancement Implementation ZMDG_BS_MAT_API_BP01_EXT Change Enhancement Implementation ZMDG_BS_MAT_API_BP01_EXT Inactive Properties History Technical Details Enh. Implementation Elements BAdI Implementation ZMDG_BS_MAT_API_BP01_IMPL Documentation Description Implementation: BAdI: Extension of the API with Customer-Specific Seg... ☐ Default Implementation ☐ Example Implementation ☐ "Active" not switchable through Custom.(IMG) Runtime Behavior ☒ Implementation is active Effect in Current Client The implementation will not be called Properties of BAdI Definition BAdI Definition Name MDG_BS_MAT_API_SEGMENTS_EXT Description BAdI: Extension of the API with Customer-Specific Segments Interface IF_MDG_BS_MAT_API_SEGMENTS_EXT Instance Creation Mode No Reuse of BAdI Instance More implementations in other systems (perhaps already deleted)										
6.	Create a new filter entry.	 Enhancement Implementation ZMDG_BS_MAT_API_BP01_EXT Change Enhancement Implementation ZMDG_BS_MAT_API_BP01_EXT Inactive Properties History Technical Details Enh. Implementation Elements BAdI Implementations Des... ▼ ZMDG_BS_MAT_API_E Impleme - Implementing Class - Filter Val. Filter Values Combination Combination Filt. Val. Filt. Val. AND OR <table><thead><tr><th>Filter Combinations</th><th>Filt...</th><th>Value 1</th><th>Com...</th><th>Filter</th></tr></thead><tbody><tr><td>▼ Combination 1</td><td></td><td>YMDGM_BUPA01_S</td><td>=</td><td>MAT_API_SEGMENT_EXT_FILTER</td></tr></tbody></table>	Filter Combinations	Filt...	Value 1	Com...	Filter	▼ Combination 1		YMDGM_BUPA01_S	=	MAT_API_SEGMENT_EXT_FILTER
Filter Combinations	Filt...	Value 1	Com...	Filter								
▼ Combination 1		YMDGM_BUPA01_S	=	MAT_API_SEGMENT_EXT_FILTER								

7.	Enter the filter value YMDGM_BUPA01_S	
----	---	--

5.1.3.1. Creating a Selection-Range Table

1.	The IS_SELECTION parameter contains the material selection criteria for which the database select must be executed. The selection criteria always contain the MATNR and possibly additional key fields of the customer-specific database table. In case you have key fields in your customer-specific table you must append MDG_BS_MAT_S_MAT_SELECTION by your key-fields. This is shown in the next step.	 <table border="1"> <thead> <tr> <th>Ty.</th> <th>Parameter</th> <th>Type spec.</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>▶</td> <td>IS_SELECTION</td> <td>TYPE MDG_BS_MAT_S_MAT_SELECTION</td> <td>MDG BS MAT: Material Selection</td> </tr> <tr> <td>▶</td> <td>ET_MESSAGE</td> <td>TYPE MDG_BS_MAT_T_MAT_MSG</td> <td>MDG BS MAT: Material Message</td> </tr> <tr> <td>▶</td> <td>ET_DATA</td> <td>TYPE ANY TABLE</td> <td>Segment Data</td> </tr> </tbody> </table> Method: IF_MDG_BS_MAT_API_SEGMENTS_EXT~READ Active	Ty.	Parameter	Type spec.	Description	▶	IS_SELECTION	TYPE MDG_BS_MAT_S_MAT_SELECTION	MDG BS MAT: Material Selection	▶	ET_MESSAGE	TYPE MDG_BS_MAT_T_MAT_MSG	MDG BS MAT: Material Message	▶	ET_DATA	TYPE ANY TABLE	Segment Data
Ty.	Parameter	Type spec.	Description															
▶	IS_SELECTION	TYPE MDG_BS_MAT_S_MAT_SELECTION	MDG BS MAT: Material Selection															
▶	ET_MESSAGE	TYPE MDG_BS_MAT_T_MAT_MSG	MDG BS MAT: Material Message															
▶	ET_DATA	TYPE ANY TABLE	Segment Data															

2.	Start transaction SE11. Open structure MDG_BS_MAT_S_MAT_SELECTION. Create an append called BUPA_ID_RANGE as shown.	 It is vital that the component name of the append adheres to the naming standard <KEY-FIELDNAME OF BACKEND TABLE>_RANGE. In our case the backend table is YMDGM_BUPA00 and the key-fieldname is BUPA_ID therefore the component name is BUPA_ID_RANGE.
3.	Here are the details for structure YBUPA_GUID_T_RANGE that you need in the previous step.	

5.1.3.2. BAdI Implementation: READ Method

1.	<pre>METHOD if_mdg_bs_mat_api_segments_ext~read. DATA: lt_mdg_bupa TYPE ymdgm_bupa01 t.</pre>
----	---

```

CLEAR: et_data, et_message.
SELECT * FROM ymdgm_bupa00 INTO CORRESPONDING FIELDS OF TABLE lt_mdgm_bupa
WHERE matnr IN is_selection-matnr_range
AND bupa_id IN is_selection-bupa_id_range.
INSERT LINES OF lt_mdgm_bupa INTO TABLE et_data.
ENDMETHOD.

```

5.1.3.3. BAdI Implementation: CHECK_AND_SAVE Method

2. METHOD if_mdgm_bs_mat_api_segments_ext~check_and_save.

```

DATA: ls_data TYPE ymdgm_bupa01_s.
DATA: ls_data_x TYPE ymdgm_bupa01_s_x.
DATA: ls_message TYPE mdgm_bs_mat_s_mat_msg.
DATA: lt_modify TYPE STANDARD TABLE OF ymdgm_bupa00.
DATA: ls_modify TYPE ymdgm_bupa00.
DATA: lt_delete TYPE STANDARD TABLE OF ymdgm_bupa00.
DATA: ls_delete TYPE ymdgm_bupa00.

CLEAR et_message.
* First some checks
LOOP AT is_data-ymdgm_bupa01_tab INTO ls_data.
  " All fields must be filled, otherwise MESSAGE e000(00) WITH ls_data-
  matnr ls_data-bupa_id ls_data-nickname.
  IF ls_data-matnr IS INITIAL OR ls_data-bupa_id IS INITIAL OR ls_data-
  nickname IS INITIAL.
    ls_message-msgid = '00'.
    ls_message-msgno = '000'.
    ls_message-msgty = 'E'.
    ls_message-msgv1 = 'Empty value detected in: '.
    ls_message-msgv2 = ls_data-matnr.
    ls_message-msgv3 = ls_data-bupa_id.
    ls_message-msgv4 = ls_data-nickname.
    INSERT ls_message INTO TABLE et_message.
  ENDIF.
  " BuPa must exist
  " ...
ENDLOOP.
* In test mode, we're finished now
CHECK iv_test_mode = abap_false.
* Determine which records to delete or to insert/update
LOOP AT is_data-ymdgm_bupa01_x_tab INTO ls_data_x.
  IF ls_data_x-delete_row = abap_true. " Keep key fields for deletion
    ls_delete-matnr = ls_data_x-matnr.
    ls_delete-bupa_id = ls_data_x-bupa_id.
    INSERT ls_delete INTO TABLE lt_delete.
  ELSE.
    READ TABLE is_data-ymdgm_bupa01_tab INTO ls_data
      WITH TABLE KEY matnr = ls_data_x-matnr
      bupa_id = ls_data_x-bupa_id.
    ASSERT sy-subrc = 0. " No _x record without data record
    MOVE-CORRESPONDING ls_data TO ls_modify.
    INSERT ls_modify INTO TABLE lt_modify.
  ENDIF.
ENDLOOP.
* Now change the DB
* Note: In a productive implementation, these DB changes must be done ON COMM

```

```

IT (PERFORM xxx ON COMMIT or CALL FUNCTION xxx IN UPDATE TASK)
* Note: We are basically ignoring the _X structure here -
  if data gets changed, all fields get updated
  IF NOT lt_delete IS INITIAL.
    DELETE ymdgm_bupa00 FROM TABLE lt_delete.
  ENDIF.
  IF NOT lt_modify IS INITIAL.
    MODIFY ymdgm_bupa00 FROM TABLE lt_modify.
  ENDIF.
ENDMETHOD.

```

5.1.3.4. BAdI Implementation: GET_ES_NODEINFO Method

Implement GET_ES_NODEINFO Method if you have extended the enterprise search model MATERIAL. See also extensibility guide for the Search: <http://scn.sap.com/docs/DOC-29002>.

With this method you get the node name in the Enterprise Search template for the given customer-defined database table with parameters ET_ES_NODENAME.

5.2. MDG Data Model Extension

In this section you will extend the MDG data model with the entity type **YBUPA01**. The entity type has one attribute **ZZNICKN** and one qualifying relationship to entity type **YBUPA**. **YBUPA01** will have two key fields **MATERIAL** and **YBUPA**. For details of the data model extension refer to the figure below.

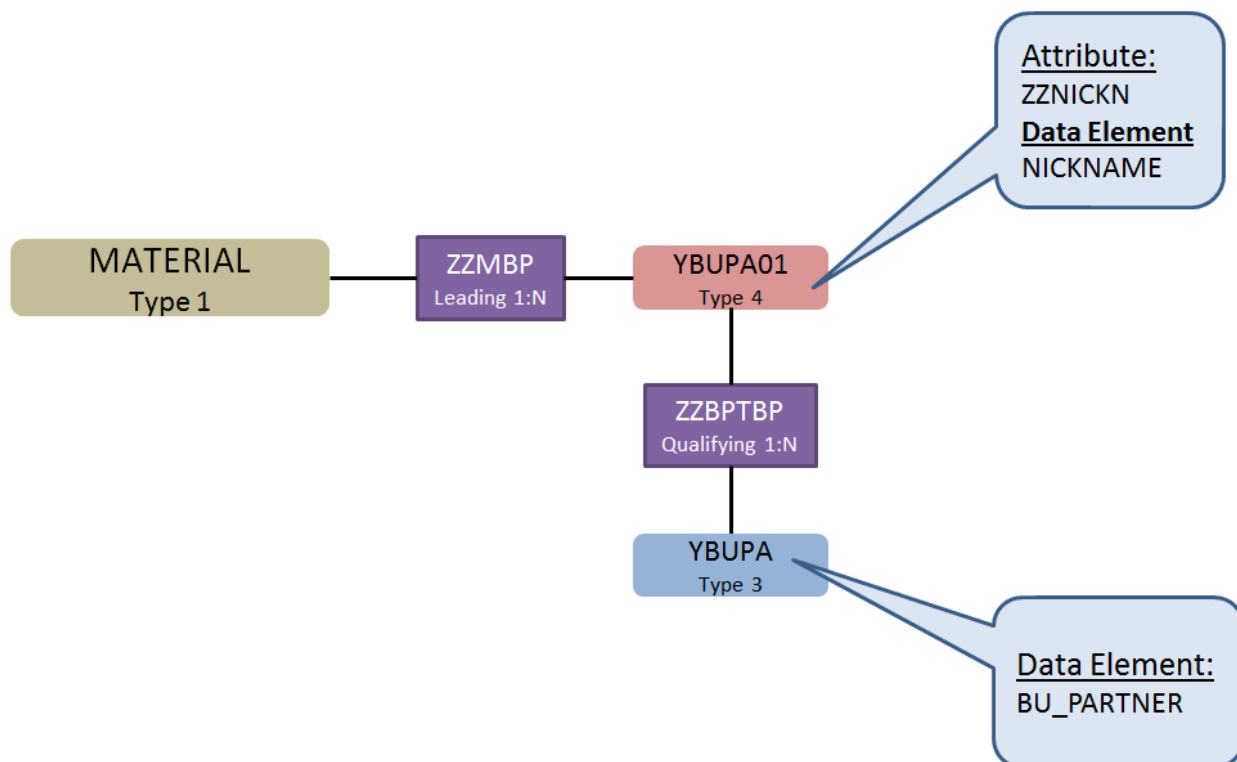
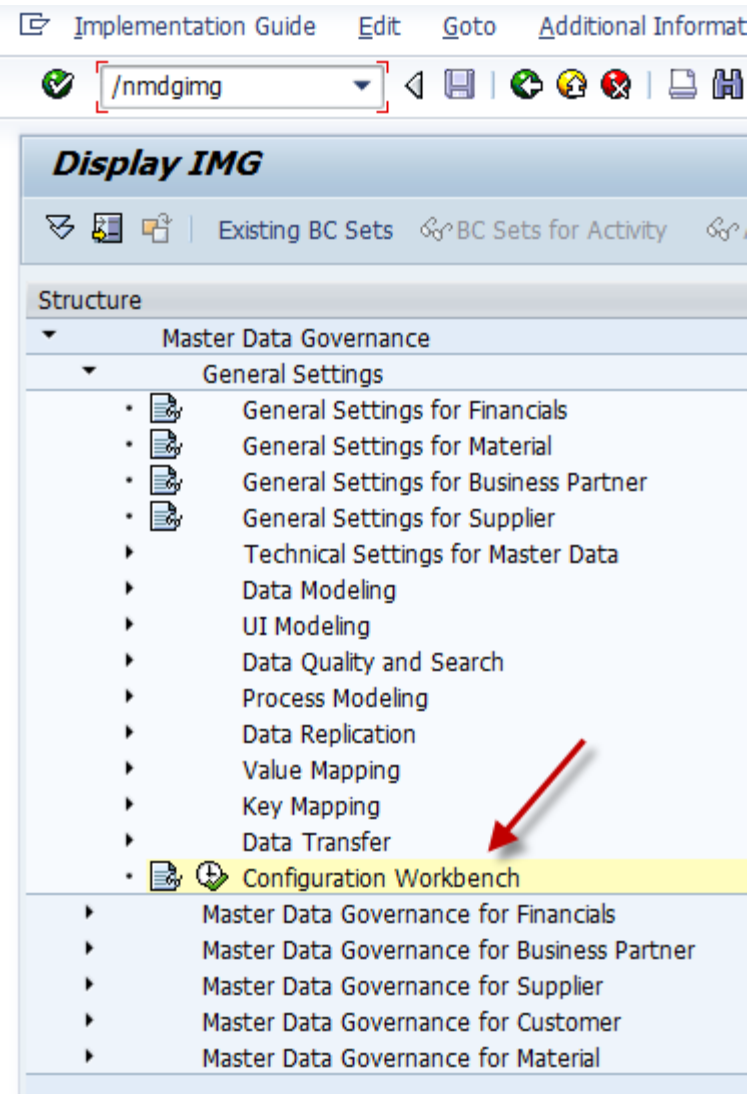


Figure: Data Model details for extension

5.2.1. Extend MDG Data Model

1. Start the MDG customizing with transaction **MDGIMG** and start the **Configuration Workbench** as shown.



2. In the Configuration Workbench open the **MM Data Model**.

Configuration Workbench

Data Models

New

Data Model	Description	Active Version
0F	Chart of Accounts	None
0G	Chart of Accounts and Org. ...	Same
BP	Business Partner	Same
MM	Material Maintenance MM01/02	Same
TD	Flight Data Model f. Field Pro...	Different
Z1	Copy of MM for message pr...	Same
Z2	Copy of MM for message pr...	Same
Z9	test	Same
ZD	SFLIGHT Test	Same
ZE	Equipment	Same
ZG	Data model for extensibility t...	Same
ZK	KP Test Data Model	Same
ZP	Business Partner	None
ZX	Test - Chart of Accounts an...	Same
ZZ	Test Data Model	Same

3. Choose the **Edit** pushbutton.

Data Model: MM, Material Maintenance MM01/02

Save and Back | Save | Back | Cancel | **Edit** | New

Edit (Ctrl+E)

▼ Data Model Details

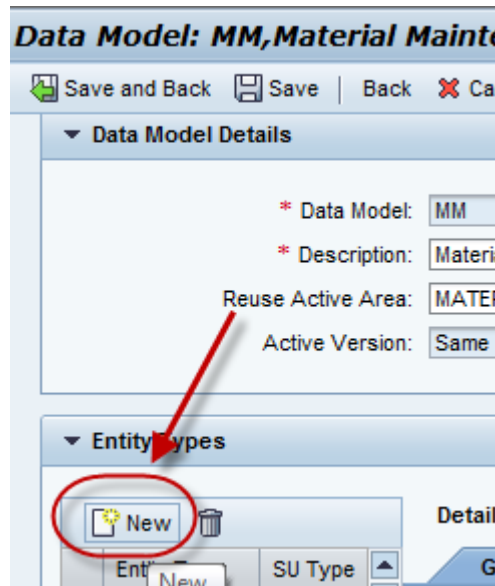
* Data Model: MM

* Description: Material Maintenance MM01/02

Reuse Active Area: MATERIAL -

Active Version: Same

4. To create a new entity type choose the **New** pushbutton.



5. Create the new entity type **YBUPA01 (SU Type 4)** with the details shown in the screenshot.

Data Model: MM, Material Maintenance MM01/02

Save and Back | Save | Back | Cancel | Edit | New | Check | Activate | Reuse Areas

Adjust Change Requests

Data Model Details

* Data Model: MM
 * Description: Material Maintenance MM01/02
 Reuse Active Area: MATERIAL -
 Active Version: Same

Changed On: 27.04.2012
 Changed At: 14:18:37
 User Name: RUETERL

Entity Types

Entity Type	SU Type
BSCDATTXT	4
CHANGENO	3
CHARD	3
CLASS	3
CLASSASGN	4
CLASSTYPE	3
EAN	3
ECOCNTR	3
GUID	3
INTCMNT	4
LANGUCODE	3
MATERIAL	1
MEAN_GTIN	4
QTEUNIT	3
UNITOFMSR	4
VALCNT	3
VALUATION	4
YMARC	4
YMARC1	4
YMARC2	4
YMARC3	4
YMARC4	4

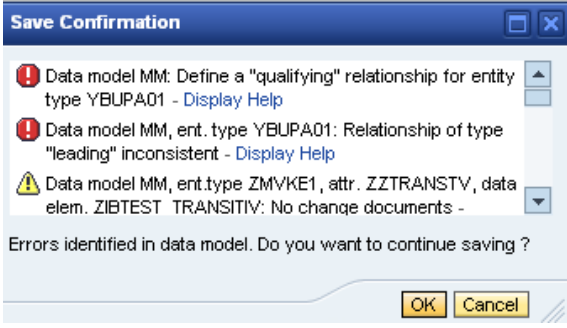
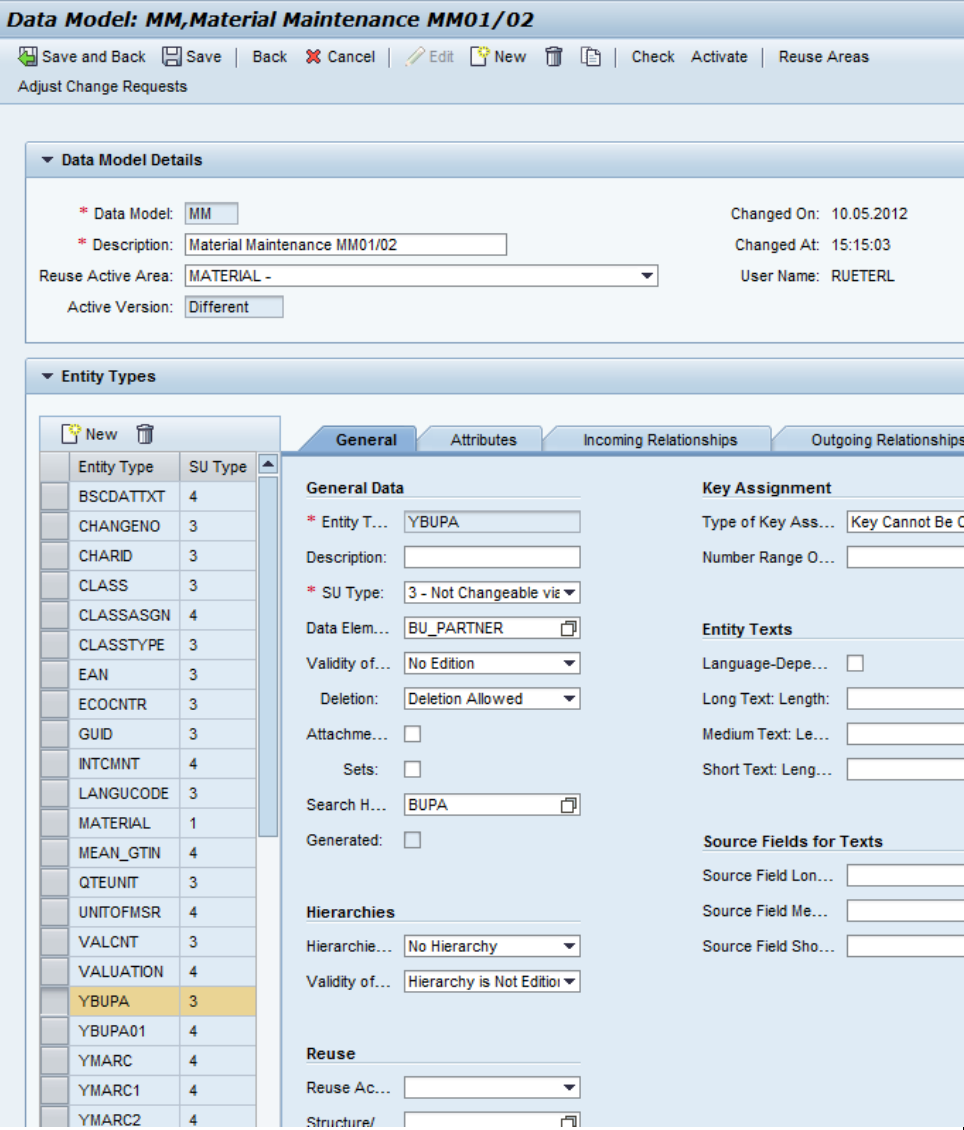
General | Attributes | Incoming Relationships | Outgoing Relationships

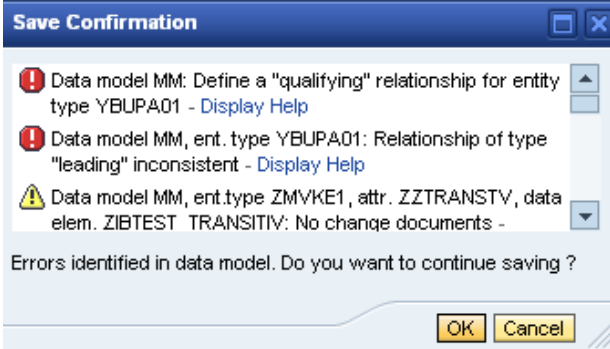
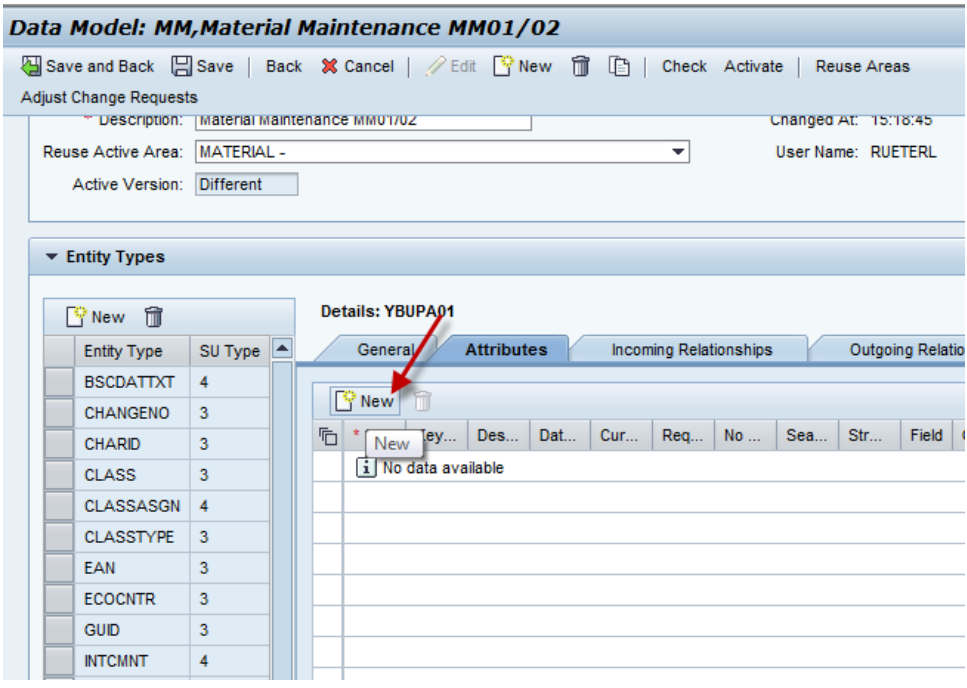
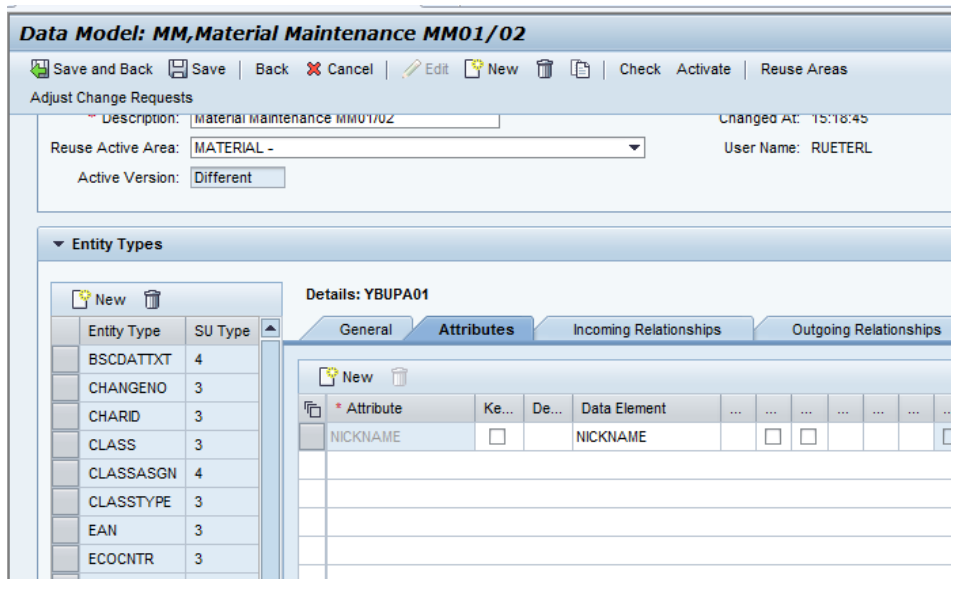
General Data

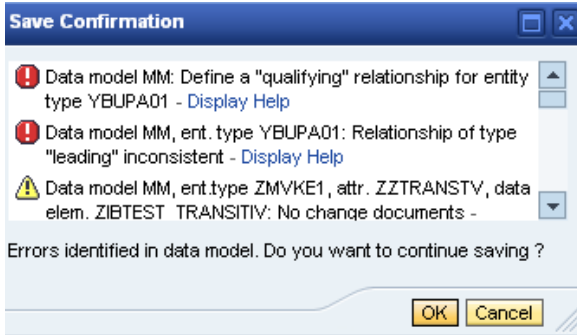
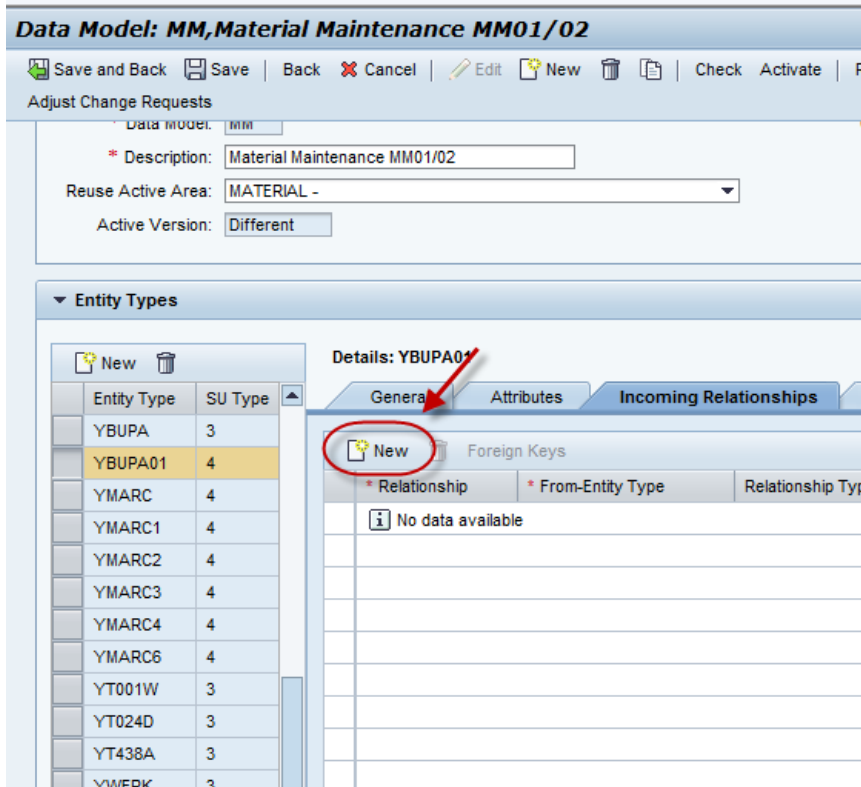
* Entity T...: YBUPA01
 Description:
 * SU Type: 4 - Changeable via Ot
 Data Elem...
 Validity of...: No Edition
 Deletion: Deletion Allowed
 Attachme...
 Sets:
 Search H...
 Generated:
Hierarchies
 Hierarchie...: No Hierarchy
 Validity of...: Hierarchy is Not Editio
Reuse
 Reuse Ac...
 Structure/...

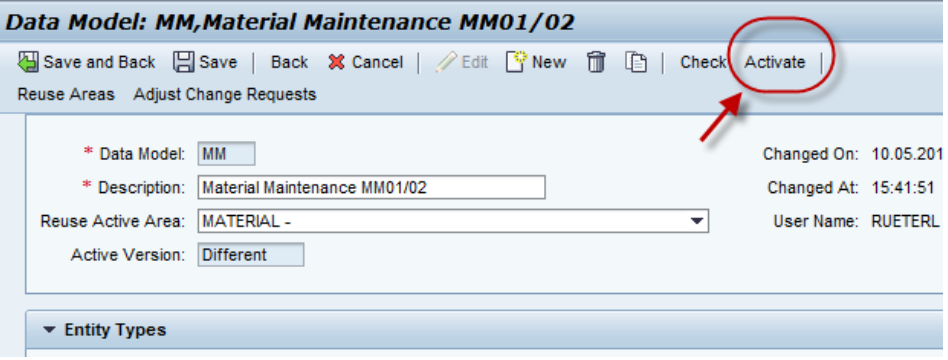
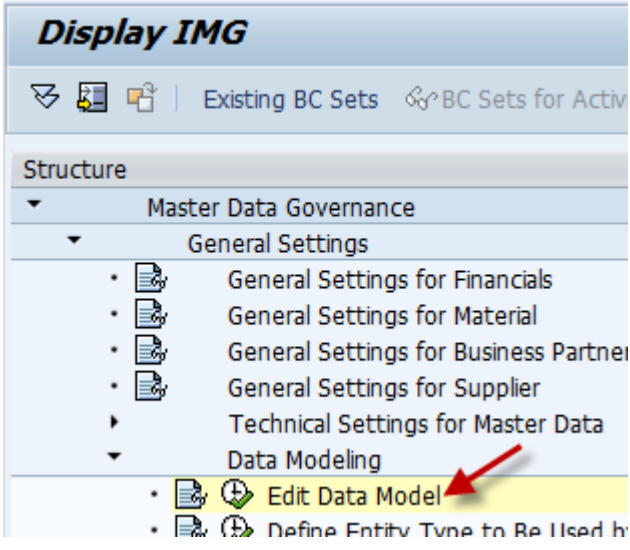
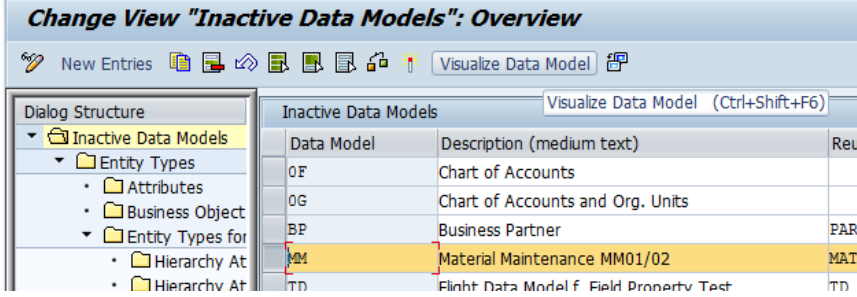
Key Assignment

Type of Key Ass...: Key Cannot Be Chang
 Number Range O...
Entity Texts
 Language-Depe...
 Long Text: Length: 00
 Medium Text: Le... 00
 Short Text: Leng... 00
Source Fields for Texts
 Source Field Lon...
 Source Field Me...
 Source Field Sho...

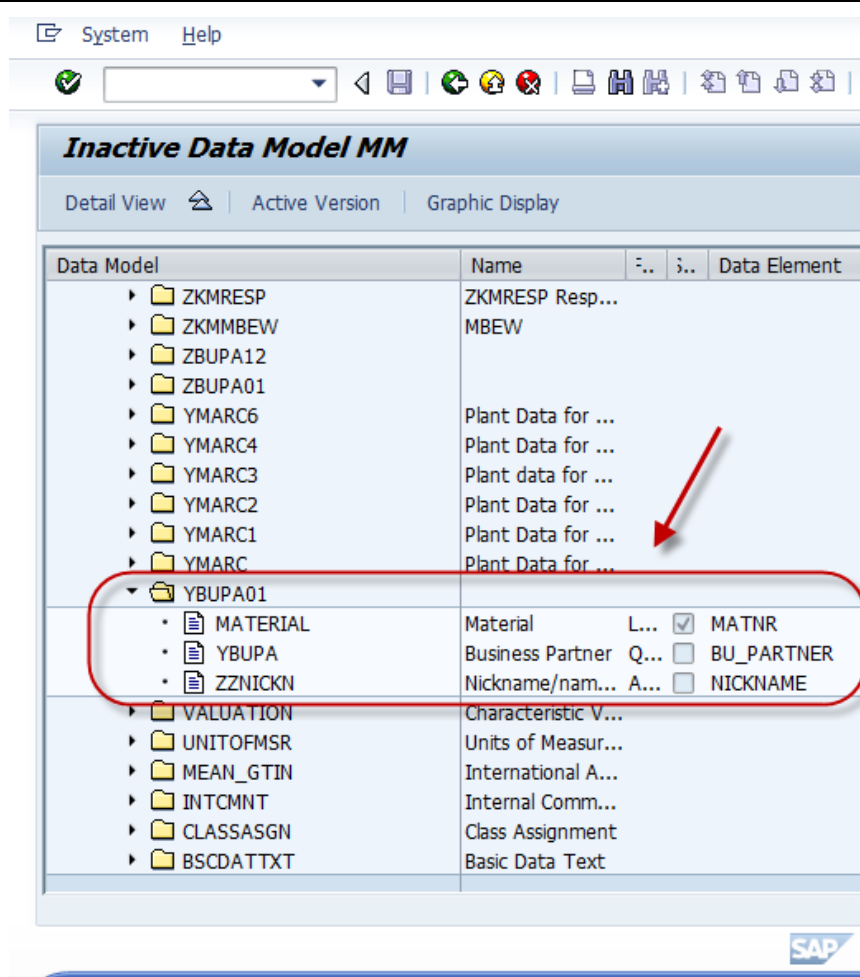
6.	Choose Save. Choose OK in the confirmation dialog.	
7.	Create the new Entity Type YBUPA (SU Type 3) with the details shown in the screenshot.	

8.	Choose Save. Choose OK in the confirmation dialog.	
9.	Go back to Entity Type YBUPA01. Switch the Attributes tab. Choose the New pushbutton.	
10.	Create the new attribute NICKNAME as shown.	

11.	Choose Save. Choose OK in the confirmation dialog.	 The dialog box titled "Save Confirmation" contains three error messages: ! Data model MM: Define a "qualifying" relationship for entity type YBUPA01 - Display Help ! Data model MM, ent. type YBUPA01: Relationship of type "leading" inconsistent - Display Help ! Data model MM, ent.type ZMVKE1, attr. ZZTRANSTV, data elem. ZIBTEST TRANSITIV: No change documents - Errors identified in data model. Do you want to continue saving ? Buttons: OK, Cancel															
12.	For Entity Type YBUPA01 open the Incoming Relationships tab. Choose the New pushbutton.	 The screenshot shows the "Data Model: MM, Material Maintenance MM01/02" interface. The "Entity Types" list on the left includes YBUPA01. The "Details: YBUPA01" panel on the right has the "Incoming Relationships" tab selected. A red circle highlights the "New" button in the "Incoming Relationships" section.															
13.	Create the two relationships as shown.	 The screenshot shows the "Details: YBUPA01" interface with the "Incoming Relationships" tab selected. The table below shows the relationships created: <table><thead><tr><th>* Relationship</th><th>* From-Entity Type</th><th>Relationship Type</th><th>Cardinality</th><th>D...</th></tr></thead><tbody><tr><td>ZZMBP</td><td>MATERIAL</td><td>Leading</td><td>1 : N</td><td></td></tr><tr><td>ZZBPTBP</td><td>YBUPA</td><td>Qualifying</td><td>1 : N</td><td></td></tr></tbody></table>	* Relationship	* From-Entity Type	Relationship Type	Cardinality	D...	ZZMBP	MATERIAL	Leading	1 : N		ZZBPTBP	YBUPA	Qualifying	1 : N	
* Relationship	* From-Entity Type	Relationship Type	Cardinality	D...													
ZZMBP	MATERIAL	Leading	1 : N														
ZZBPTBP	YBUPA	Qualifying	1 : N														

14.	Choose the Activate pushbutton to activate your data model.																			
15.	To check the activated data model open the Edit Data Model Customizing activity as shown.																			
16.	Choose the Visualize Data Model pushbutton .	 <table border="1" data-bbox="820 1249 1404 1417"> <thead> <tr> <th>Data Model</th> <th>Description (medium text)</th> <th>Reu</th> </tr> </thead> <tbody> <tr> <td>0F</td> <td>Chart of Accounts</td> <td></td> </tr> <tr> <td>0G</td> <td>Chart of Accounts and Org. Units</td> <td></td> </tr> <tr> <td>BP</td> <td>Business Partner</td> <td>PAR</td> </tr> <tr> <td>MM</td> <td>Material Maintenance MM01/02</td> <td>MAT</td> </tr> <tr> <td>TD</td> <td>Flight Data Model f. Field Property Test</td> <td>TD</td> </tr> </tbody> </table>	Data Model	Description (medium text)	Reu	0F	Chart of Accounts		0G	Chart of Accounts and Org. Units		BP	Business Partner	PAR	MM	Material Maintenance MM01/02	MAT	TD	Flight Data Model f. Field Property Test	TD
Data Model	Description (medium text)	Reu																		
0F	Chart of Accounts																			
0G	Chart of Accounts and Org. Units																			
BP	Business Partner	PAR																		
MM	Material Maintenance MM01/02	MAT																		
TD	Flight Data Model f. Field Property Test	TD																		

17. Make sure that your extension of the MDG Data Model looks similar to the details shown in the screenshot.

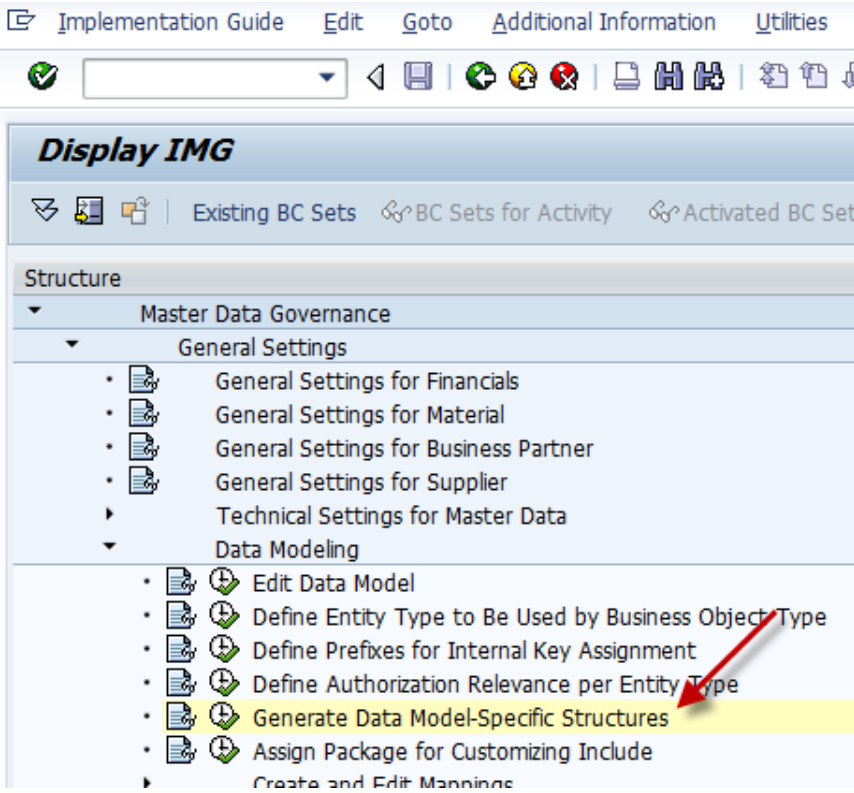
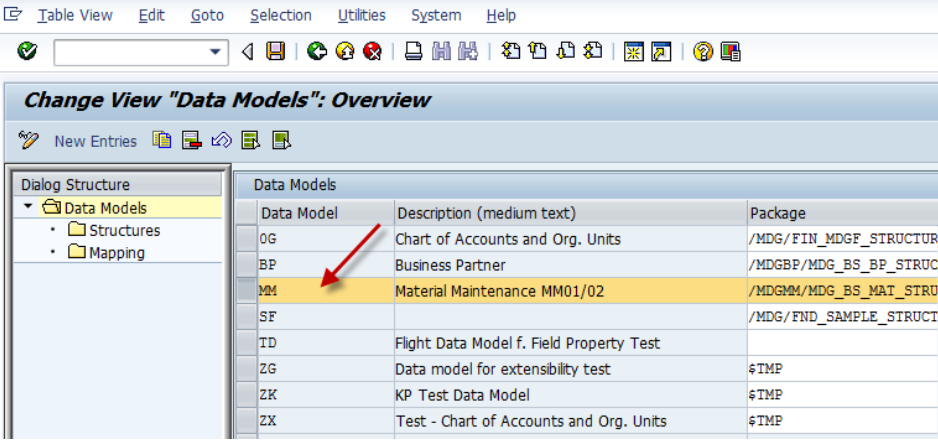


5.2.2. Generate Model-Specific Structures

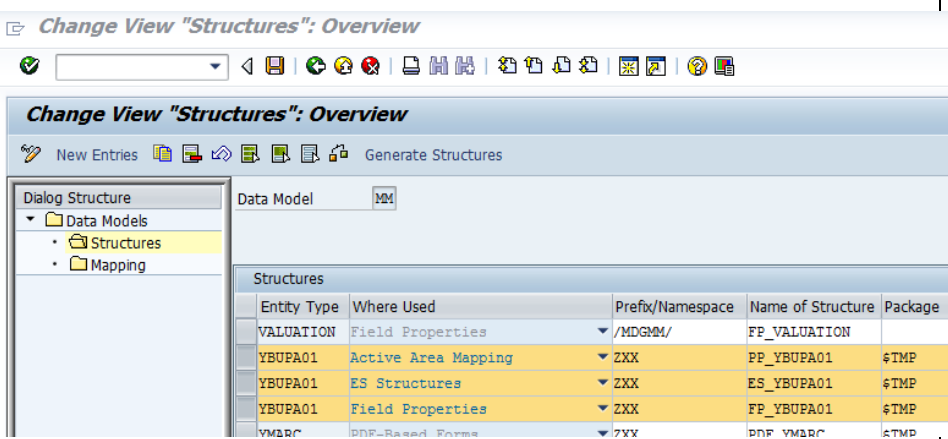
Every time you change the MDG data model, you must regenerate the structures. In this Customizing activity, for each data model and entity type you generate technical structures in the ABAP Dictionary. The system uses these structures internally for implementing the staging area. To generate these Data Model-Specific structures follow the steps below.

Note

In general if you change a data model (for example, if you change attributes of entity types or relationships); you must regenerate the structures.

1.	In MDG customizing start Generate Data Model-Specific Structures .																												
2.	Select Data Model MM . Open the Structures view.	 <table><thead><tr><th>Data Model</th><th>Description (medium text)</th><th>Package</th></tr></thead><tbody><tr><td>OG</td><td>Chart of Accounts and Org. Units</td><td>/MDG/FIN_MDGF_STRUCTURE</td></tr><tr><td>BP</td><td>Business Partner</td><td>/MDGBP/MDG_BS_BP_STRUC</td></tr><tr><td>MM</td><td>Material Maintenance MM01/02</td><td>/MDGMM/MDG_BS_MAT_STRU</td></tr><tr><td>SF</td><td></td><td>/MDG/FND_SAMPLE_STRUC</td></tr><tr><td>TD</td><td>Flight Data Model f. Field Property Test</td><td></td></tr><tr><td>ZG</td><td>Data model for extensibility test</td><td>\$TMP</td></tr><tr><td>ZK</td><td>KP Test Data Model</td><td>\$TMP</td></tr><tr><td>ZX</td><td>Test - Chart of Accounts and Org. Units</td><td>\$TMP</td></tr></tbody></table>	Data Model	Description (medium text)	Package	OG	Chart of Accounts and Org. Units	/MDG/FIN_MDGF_STRUCTURE	BP	Business Partner	/MDGBP/MDG_BS_BP_STRUC	MM	Material Maintenance MM01/02	/MDGMM/MDG_BS_MAT_STRU	SF		/MDG/FND_SAMPLE_STRUC	TD	Flight Data Model f. Field Property Test		ZG	Data model for extensibility test	\$TMP	ZK	KP Test Data Model	\$TMP	ZX	Test - Chart of Accounts and Org. Units	\$TMP
Data Model	Description (medium text)	Package																											
OG	Chart of Accounts and Org. Units	/MDG/FIN_MDGF_STRUCTURE																											
BP	Business Partner	/MDGBP/MDG_BS_BP_STRUC																											
MM	Material Maintenance MM01/02	/MDGMM/MDG_BS_MAT_STRU																											
SF		/MDG/FND_SAMPLE_STRUC																											
TD	Flight Data Model f. Field Property Test																												
ZG	Data model for extensibility test	\$TMP																											
ZK	KP Test Data Model	\$TMP																											
ZX	Test - Chart of Accounts and Org. Units	\$TMP																											
3.	Create the tree entries in the structures table for entity type YBUPA01 .	<table><tr><td>YBUPA01</td><td>Active Area Mapping</td><td>ZXX</td><td>PP_YBUPA01</td><td>\$TMP</td></tr><tr><td>YBUPA01</td><td>ES Structures</td><td>ZXX</td><td>ES_YBUPA01</td><td>\$TMP</td></tr><tr><td>YBUPA01</td><td>Field Properties</td><td>ZXX</td><td>FP_YBUPA01</td><td>\$TMP</td></tr></table>	YBUPA01	Active Area Mapping	ZXX	PP_YBUPA01	\$TMP	YBUPA01	ES Structures	ZXX	ES_YBUPA01	\$TMP	YBUPA01	Field Properties	ZXX	FP_YBUPA01	\$TMP												
YBUPA01	Active Area Mapping	ZXX	PP_YBUPA01	\$TMP																									
YBUPA01	ES Structures	ZXX	ES_YBUPA01	\$TMP																									
YBUPA01	Field Properties	ZXX	FP_YBUPA01	\$TMP																									

4. Save your changes.



Entity Type	Where Used	Prefix/Namespace	Name of Structure	Package
VALUATION	Field Properties	/MDGMM/	FP_VALUATION	
YBUPA01	Active Area Mapping	ZXX	FP_YBUPA01	\$TMP
YBUPA01	ES Structures	ZXX	ES_YBUPA01	\$TMP
YBUPA01	Field Properties	ZXX	FP_YBUPA01	\$TMP
YMARC	PDF-Based Forms	ZXX	PDF_YMARC	\$TMP

5.2.1. Clear UI Metadata Buffers

After finishing the UI, clear the metadata buffers. You can find the report in customizing Master Data Governance, Central Governance-> Master Data Governance for Material-> Clear UI Metadata Buffers.

This report clears the following buffers in this sequence:

- Text Buffer
- Search Help Buffer
- SMT-Mapping Data Buffer
- SPI Metadata Buffer

You should use this report after extending the data model to make sure that the metadata is consistent with the MDG customizing and UI configuration.

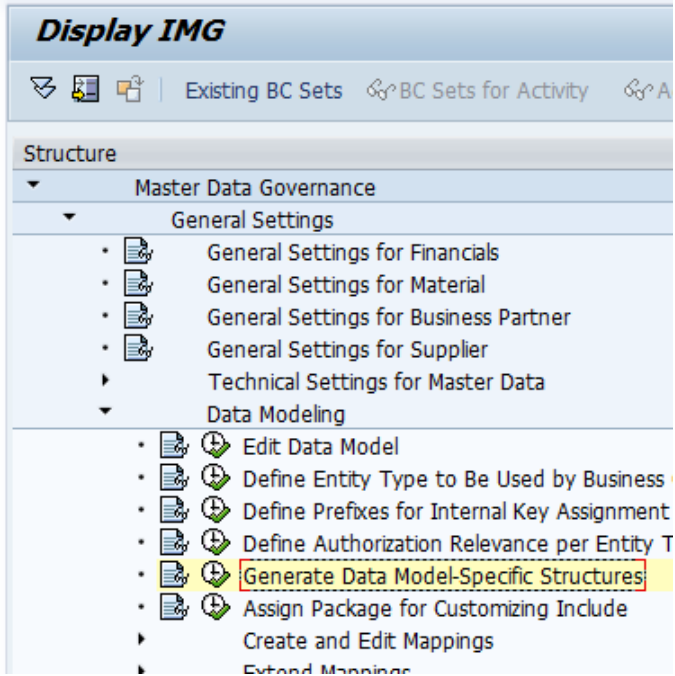
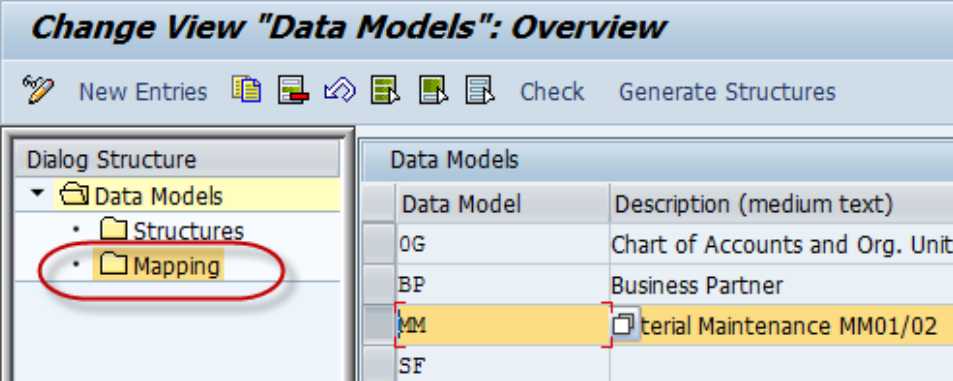
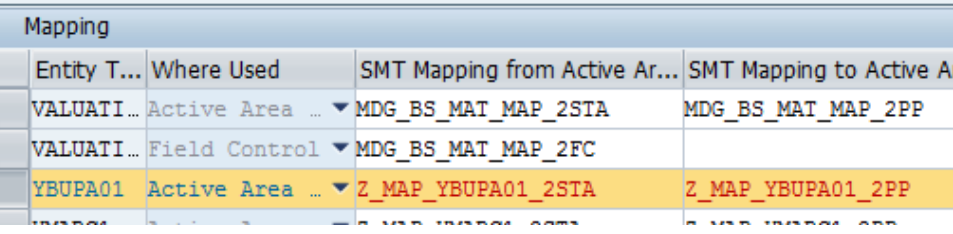
5.3. Create SMT-Mapping

You extend mappings by creating new transformations (complex transformations, field mappings) and field checks for them or by editing them.

Important

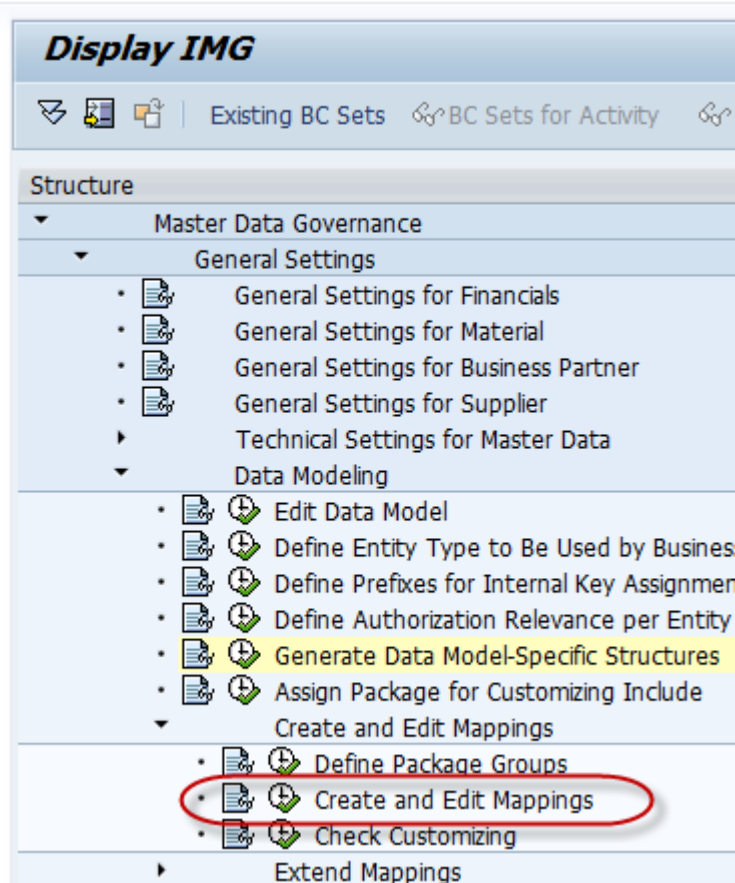
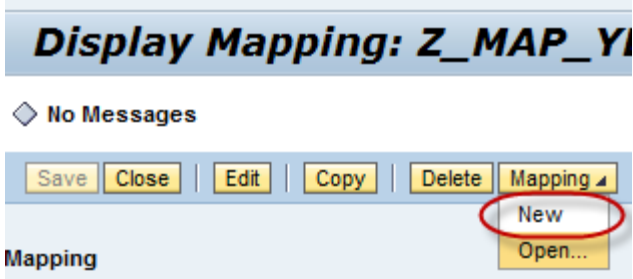
When the mappings are saved the corresponding coding is generated. Make sure that all relevant structures are ready before you start.

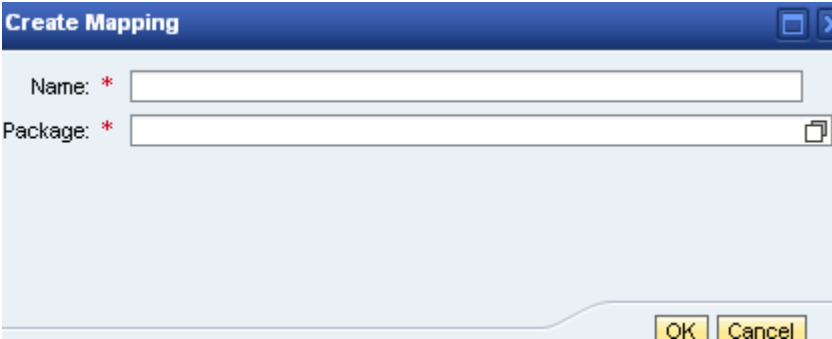
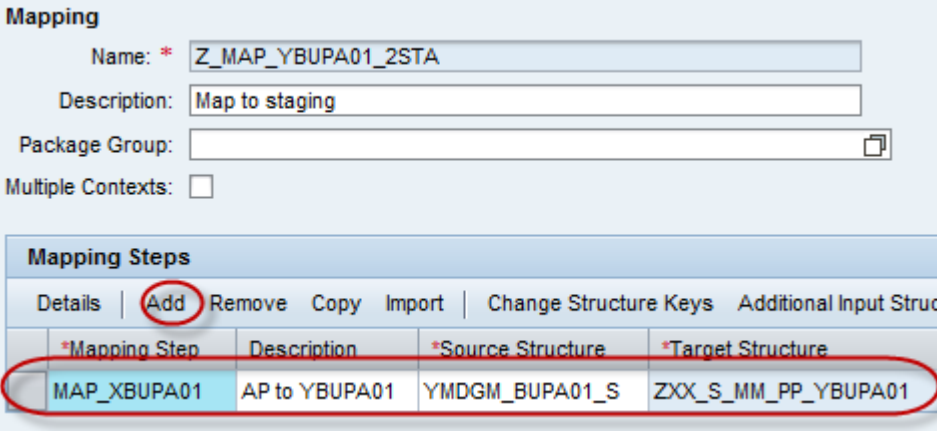
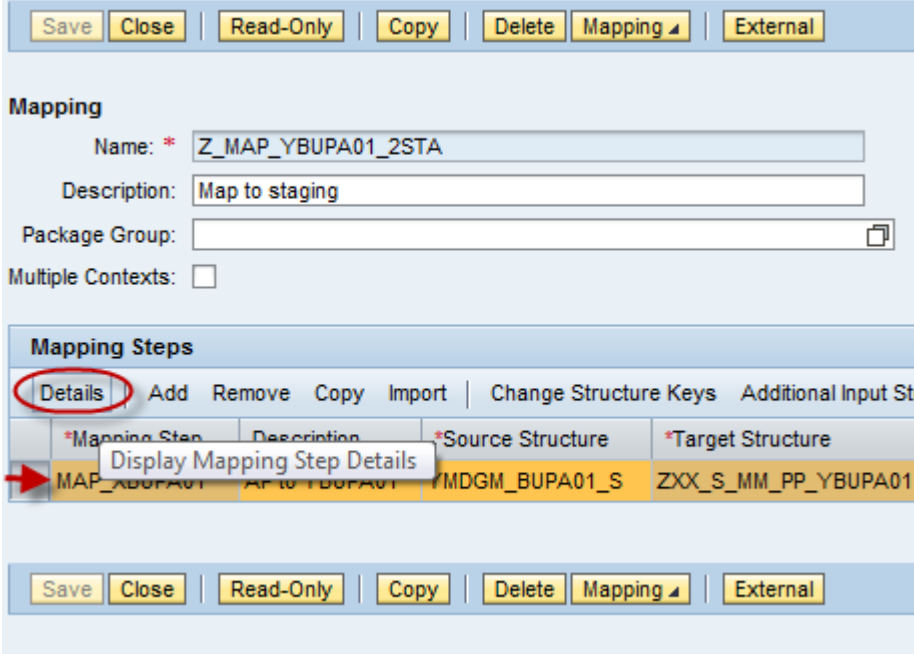
5.3.1. Create Mapping Entries in Customizing

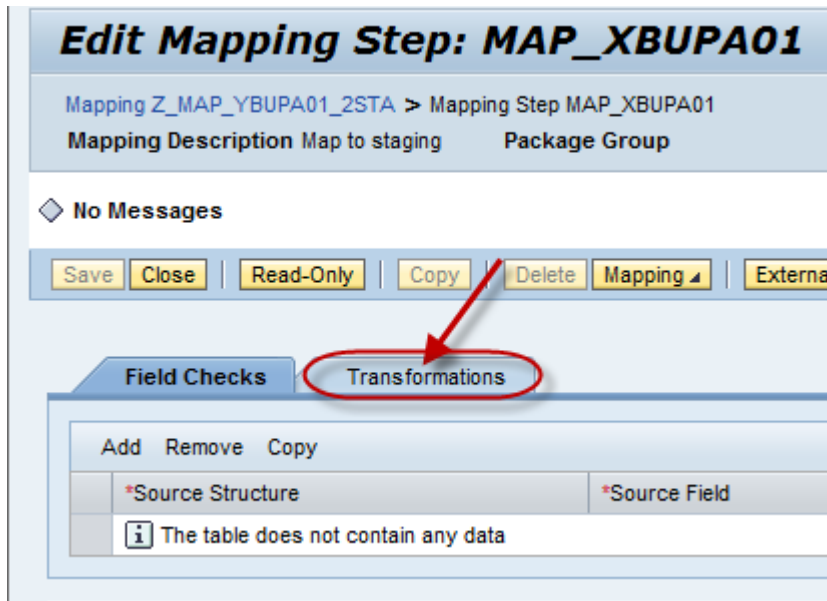
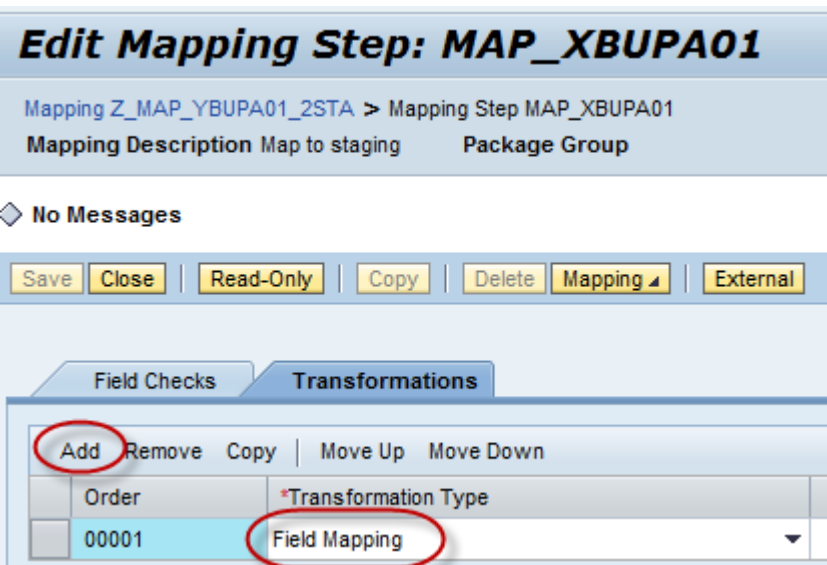
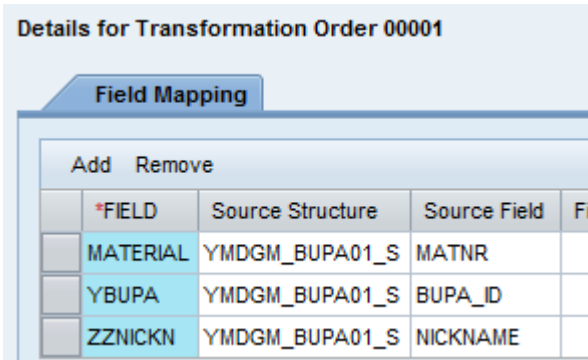
1.	In MDG customizing start Generate Data Model-Specific Structures .	
2.	Select the MM data model. Open the Mapping view.	
3.	Create a new entry for Entity Type YBUPA01 / Active Area Mapping. Mapping from the active area to the staging area: Z_MAP_YBUPA01_2STA Mapping from the staging area to the active area: Z_MAP_YBUPA01_2	

	PP	
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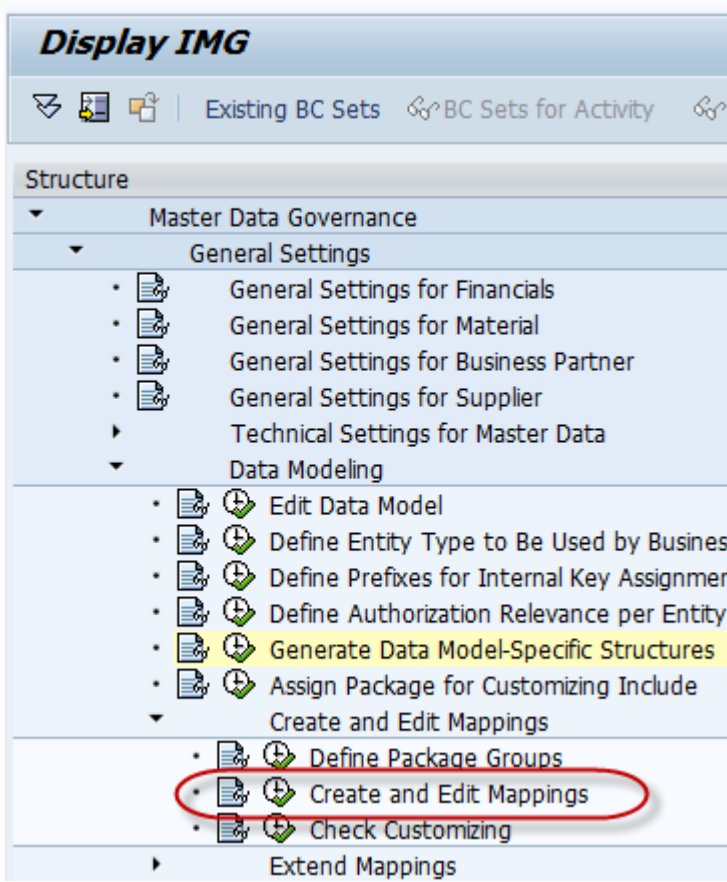
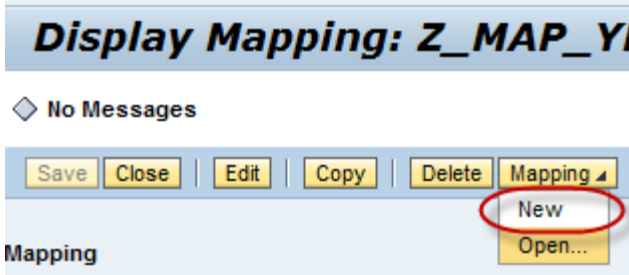
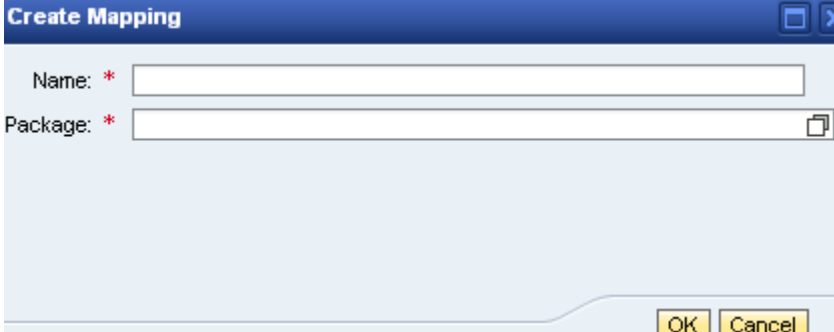
5.3.2.Map the Active Area to the Staging Area

4.	In MDG customizing start Create and Edit Mappings .	 The screenshot shows the SAP MDG Customizing menu. The path is: Master Data Governance > General Settings > Data Modeling > Create and Edit Mappings. The 'Create and Edit Mappings' option is circled in red.
5.	Select Mapping -> New .	 The screenshot shows the SAP MDG Mapping screen titled 'Display Mapping: Z_MAP_YI'. The 'Mapping' button is highlighted, and the 'New' option in the dropdown menu is circled in red.

6.	Enter the name of the mapping: Z_MAP_YBUPA01_2STA									
7.	Create a new mapping step as shown.	 <table><thead><tr><th>*Mapping Step</th><th>Description</th><th>*Source Structure</th><th>*Target Structure</th></tr></thead><tbody><tr><td>MAP_XBUPA01</td><td>AP to YBUPA01</td><td>YMDGM_BUPA01_S</td><td>ZXX_S_MM_PP_YBUPA01</td></tr></tbody></table>	*Mapping Step	Description	*Source Structure	*Target Structure	MAP_XBUPA01	AP to YBUPA01	YMDGM_BUPA01_S	ZXX_S_MM_PP_YBUPA01
*Mapping Step	Description	*Source Structure	*Target Structure							
MAP_XBUPA01	AP to YBUPA01	YMDGM_BUPA01_S	ZXX_S_MM_PP_YBUPA01							
8.	Select you mapping step and choose the Details pushbutton.	 <table><thead><tr><th>*Mapping Step</th><th>Description</th><th>*Source Structure</th><th>*Target Structure</th></tr></thead><tbody><tr><td>MAP_XBUPA01</td><td>AP to YBUPA01</td><td>YMDGM_BUPA01_S</td><td>ZXX_S_MM_PP_YBUPA01</td></tr></tbody></table>	*Mapping Step	Description	*Source Structure	*Target Structure	MAP_XBUPA01	AP to YBUPA01	YMDGM_BUPA01_S	ZXX_S_MM_PP_YBUPA01
*Mapping Step	Description	*Source Structure	*Target Structure							
MAP_XBUPA01	AP to YBUPA01	YMDGM_BUPA01_S	ZXX_S_MM_PP_YBUPA01							

9.	Open the Transformations tab.																					
10.	Add a new Field Mapping .																					
11.	Enter the field mappings as shown. Save your changes.	 <table><thead><tr><th></th><th>*FIELD</th><th>Source Structure</th><th>Source Field</th><th>Field</th></tr></thead><tbody><tr><td></td><td>MATERIAL</td><td>YMDGM_BUPA01_S</td><td>MATNR</td><td></td></tr><tr><td></td><td>YBUPA</td><td>YMDGM_BUPA01_S</td><td>BUPA_ID</td><td></td></tr><tr><td></td><td>ZZNICKN</td><td>YMDGM_BUPA01_S</td><td>NICKNAME</td><td></td></tr></tbody></table>		*FIELD	Source Structure	Source Field	Field		MATERIAL	YMDGM_BUPA01_S	MATNR			YBUPA	YMDGM_BUPA01_S	BUPA_ID			ZZNICKN	YMDGM_BUPA01_S	NICKNAME	
	*FIELD	Source Structure	Source Field	Field																		
	MATERIAL	YMDGM_BUPA01_S	MATNR																			
	YBUPA	YMDGM_BUPA01_S	BUPA_ID																			
	ZZNICKN	YMDGM_BUPA01_S	NICKNAME																			

5.3.3.Map the Staging Area to the Active Area

1.	In MDG customizing start Create and Edit Mappings .	
2.	Select Mapping -> New to create the next mapping.	
3.	Enter the following mapping name: Z_MAP_YBUPA01_2 PP	

4. Choose **Add** to create new mapping step using the details shown in the screenshot.

Edit Mapping: Z_MAP_YBUPA01_2PP

◆ No Messages

Save Close Read-Only Copy Delete Mapping External

Mapping

Name: * Z_MAP_YBUPA01_2PP

Description:

Package Group:

Multiple Contexts: ☐

Mapping Steps

Details Add Remove Copy Import Change Structure Keys Additional Input Structures Assign Service Inf

Mapping Step	Description	*Source Structure	*Target Structure	Change Structure	Change Structure Keys Exist
MAP_ZBUPA01	map YBUPA01	ZXX_S_MM_PP_YBUPA01	YMDGM_BUPA01_S	YMDGM_BUPA01_S_X	☒

Save Close Read-Only Copy Delete Mapping External

5. Choose the **Change Structure Keys** pushbutton. Then add the key fields of the change structure, as shown.

Edit Mapping: Z_MAP_YBUPA01_2PP

◆ No Messages

Save Close Read-Only Copy Delete Mapping External

Mapping

Name: * Z_MAP_YBUPA01_2PP

Description:

Package Group:

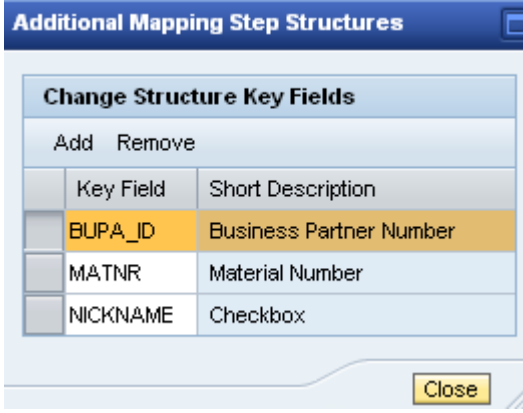
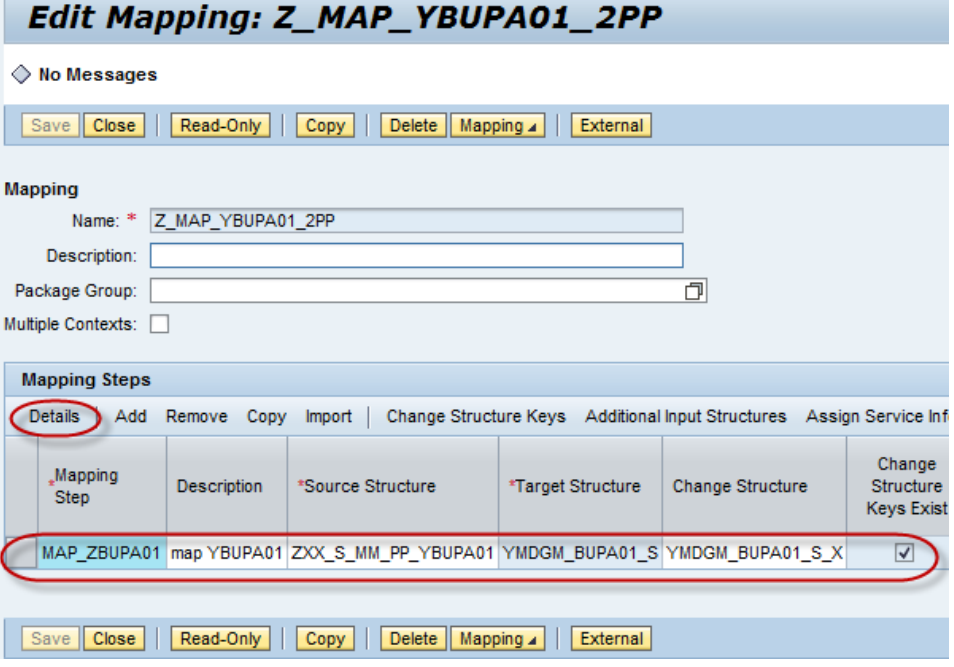
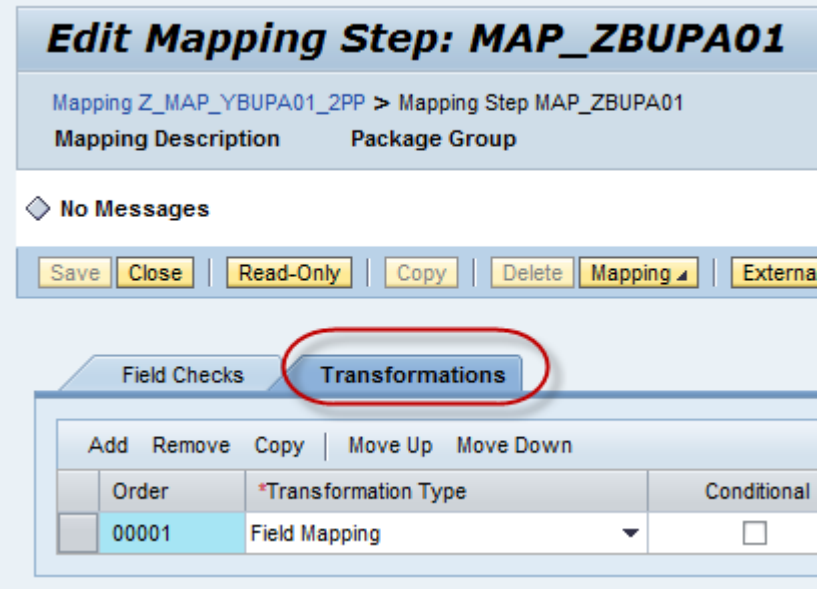
Multiple Contexts: ☐

Mapping Steps

Details Add Remove Copy Import Change Structure Keys Additional Input Structures Assign Service Inf

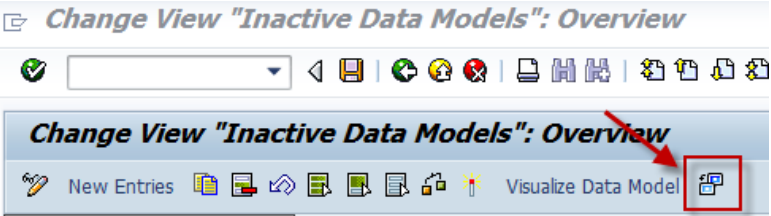
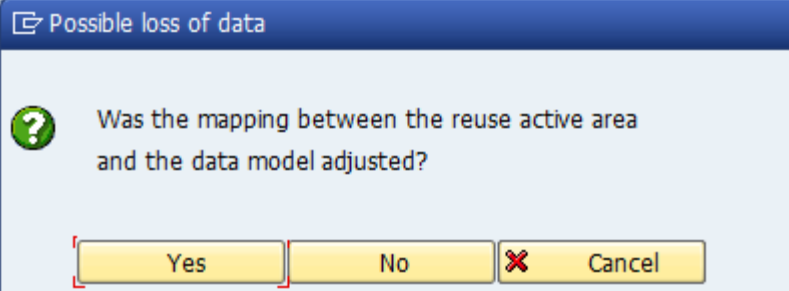
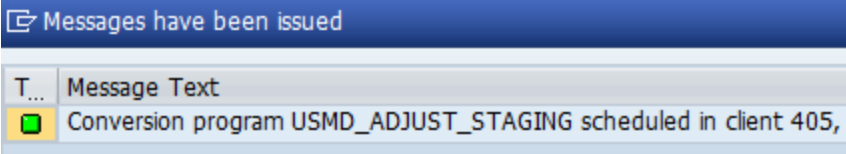
Mapping Step	Description	*Source Structure	*Target Structure	Change Structure	Change Structure Keys Exist
MAP_ZBUPA01	map YBUPA01	ZXX_S_MM_PP_YBUPA01	YMDGM_BUPA01_S	YMDGM_BUPA01_S_X	☒

Save Close Read-Only Copy Delete Mapping External

6.	Add the key field BUPA_ID as shown.	
7.	Select your mapping step and choose the Details pushbutton.	
8.	Open the Transformations tab.	

5.4. Adjust Staging Area of Linked Change Requests

This step is necessary to adjust any open change requests after you have changed the data model.

1.	Start Customizing for Master Data Governance (transaction MDGIMG). Go to General Settings -> Data Modeling -> Edit Data Model. Select data model MM. Double click on Entity Types. Choose the pushbutton Adjust staging area of linked change requests	
2.	Choose the Yes pushbutton.	
3.	The following message appears. Note: Make sure that user DDIC exist in all relevant clients.	

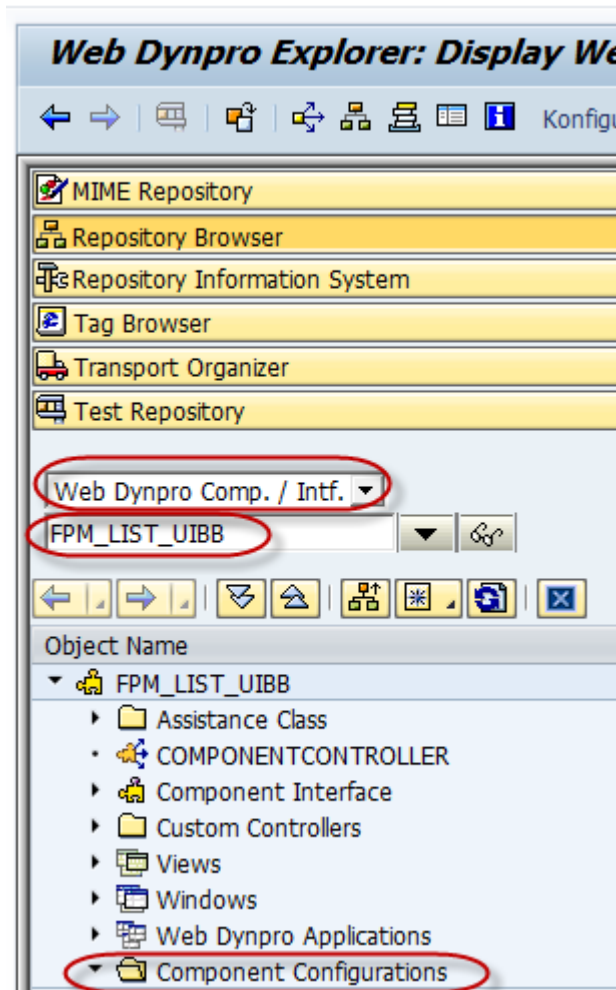
5.5. Extend User Interface

You can follow the described example or you can create an ATS list UIBB instead without a template.

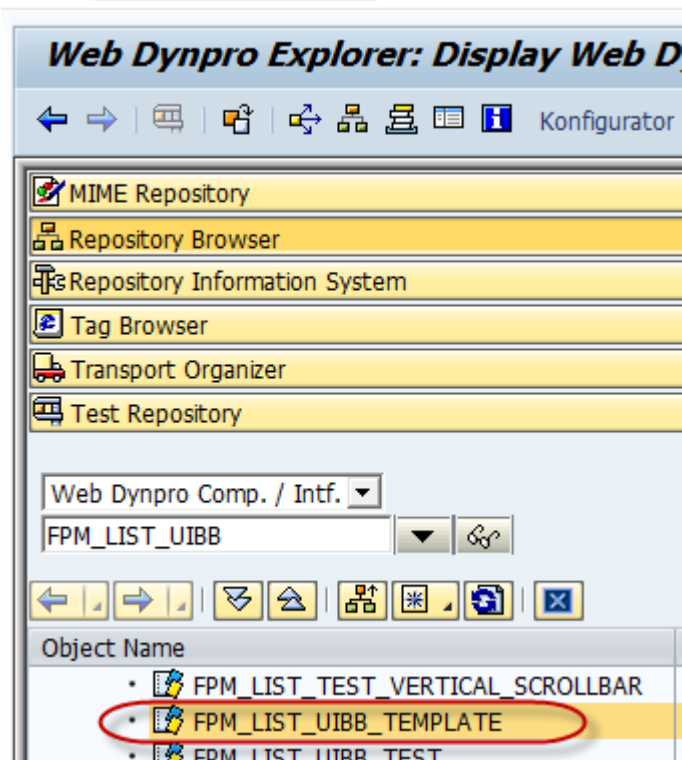
See also extensibility guide for UI: <http://scn.sap.com/docs/DOC-30192>.

5.5.1. Create FPM List UIBB

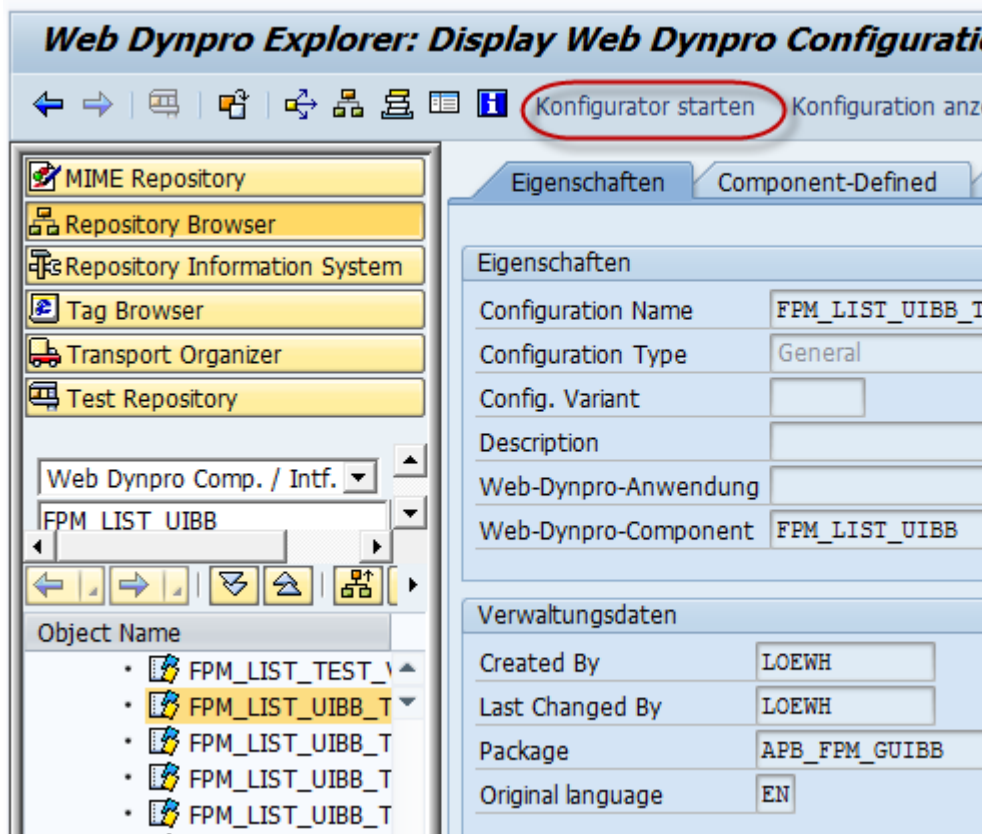
1. Navigate to the Component Configuration as shown in the screenshot.

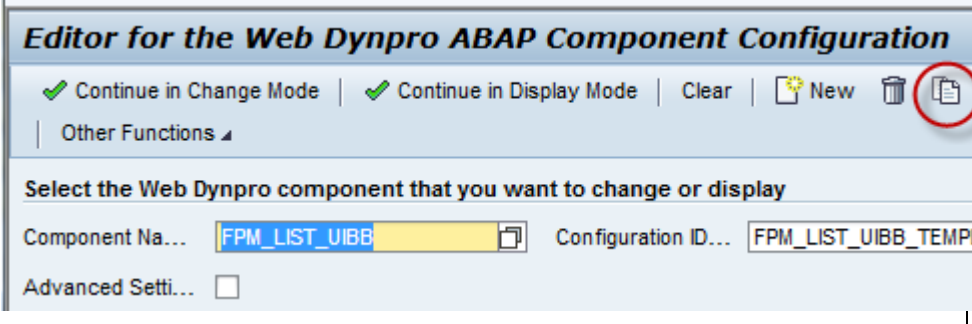
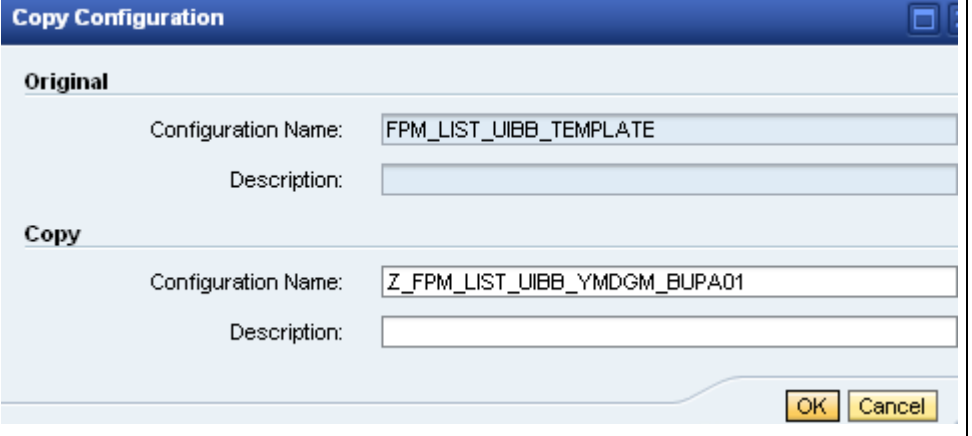


2. Open configuration
FPM_LIST_UIBB_T
EMPLATE.

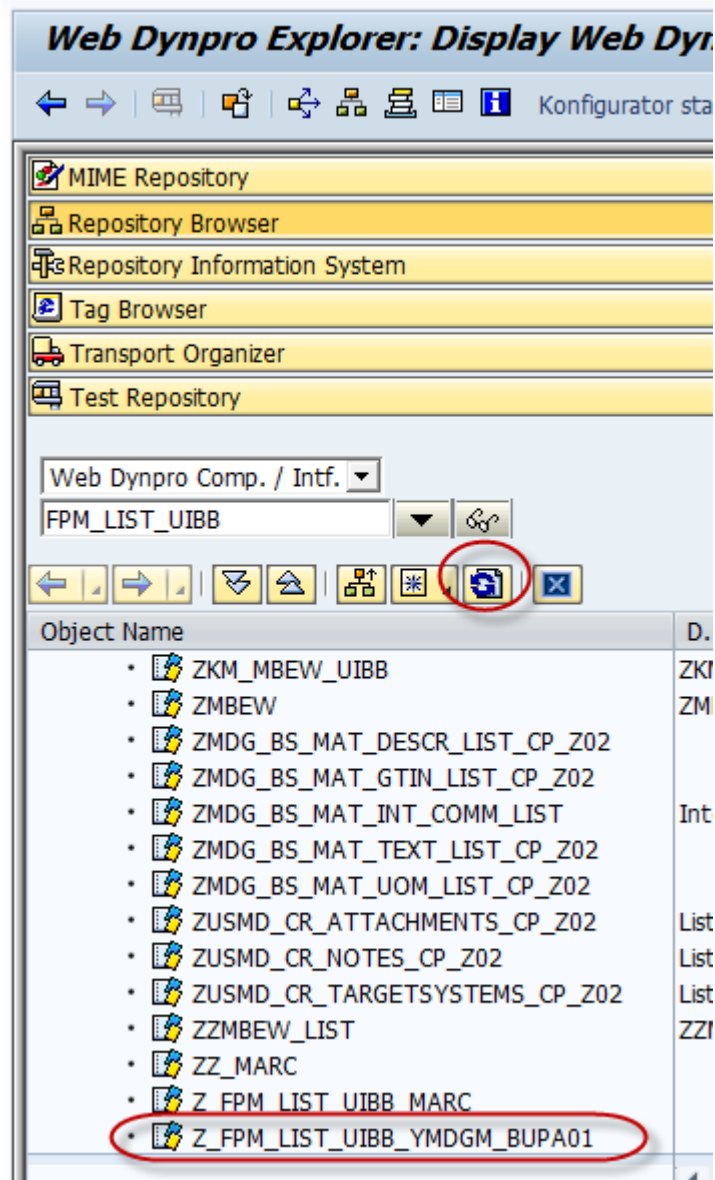


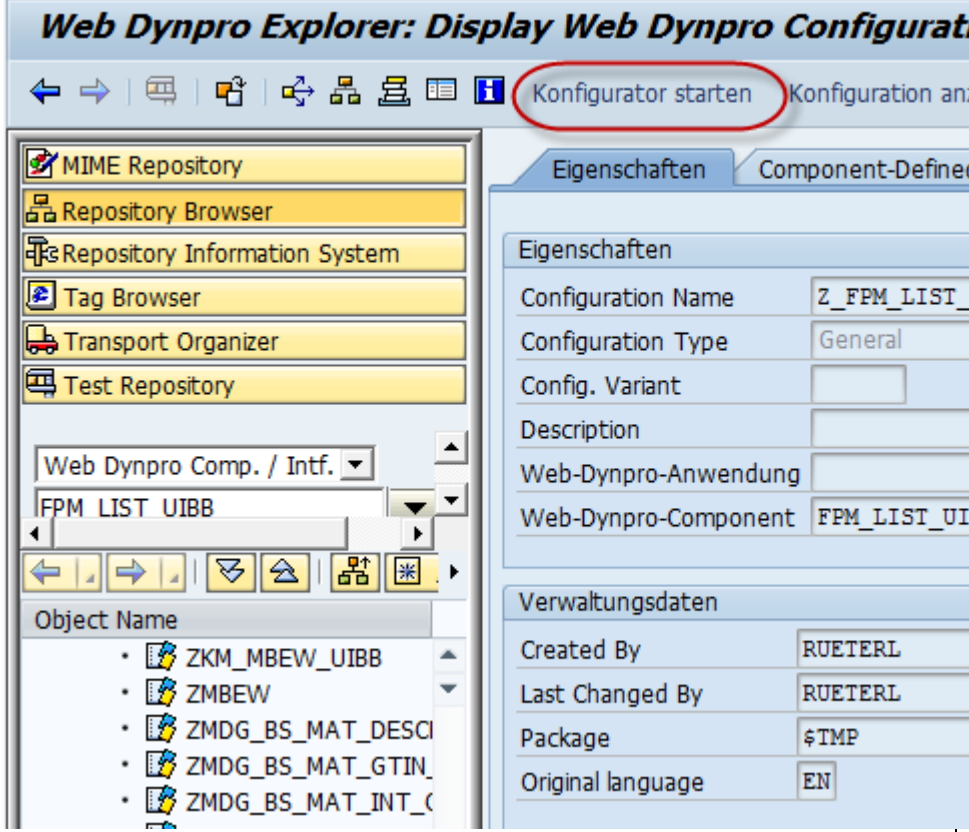
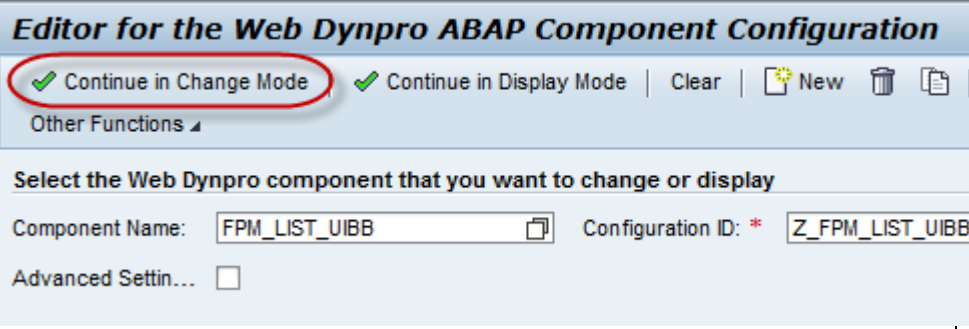
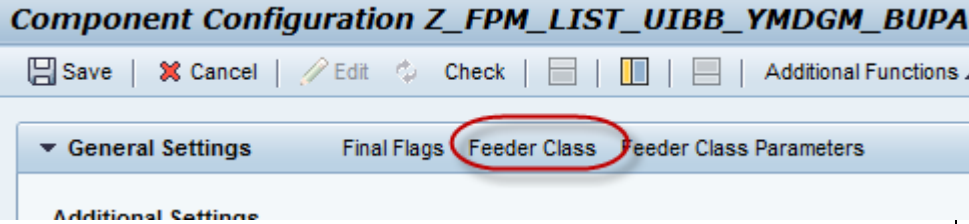
3. Choose the **Start Configuration** pushbutton.

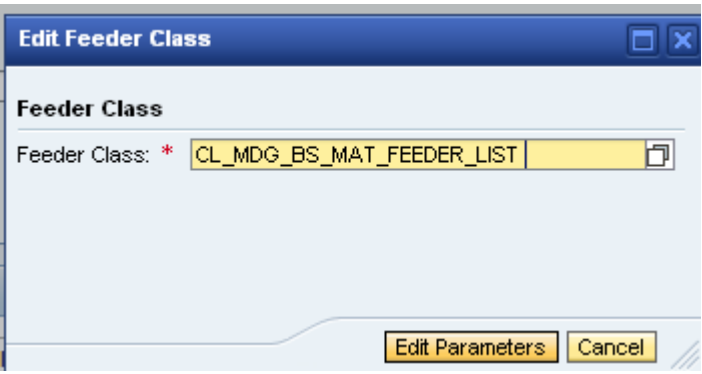
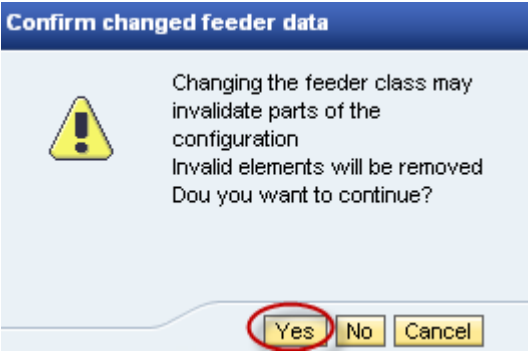
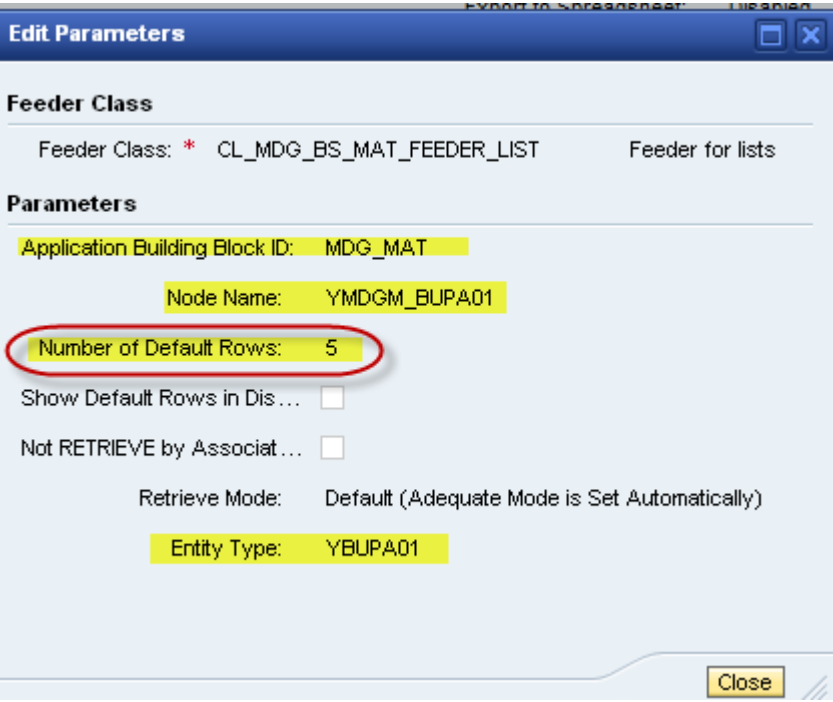


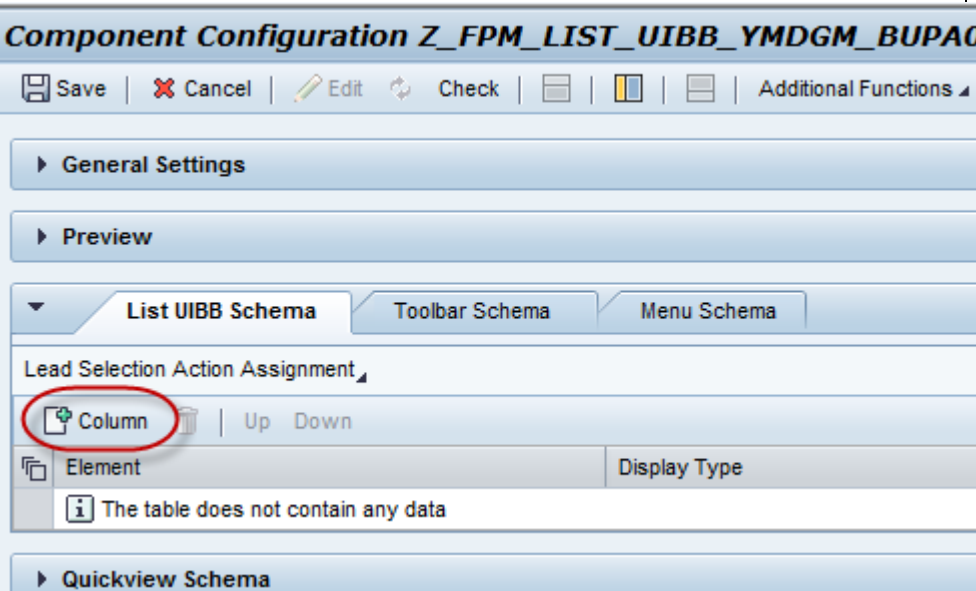
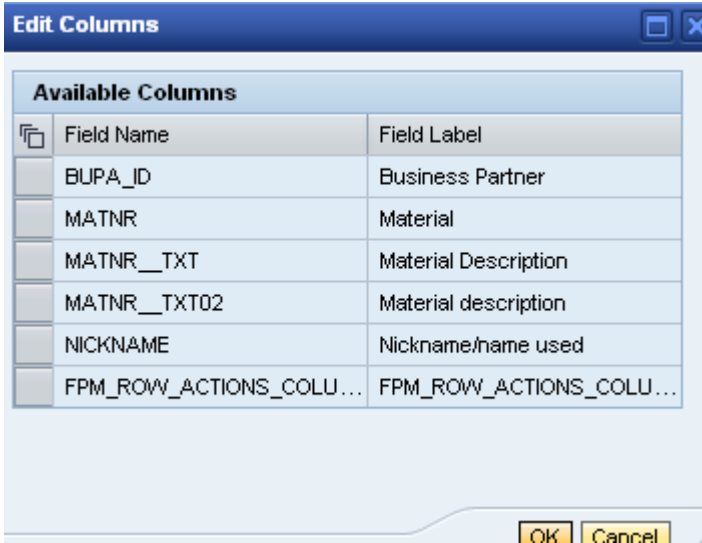
4.	Choose the Copy icon.	
5.	Enter a name for the copy as shown.	

6. Refresh the navigation tree and open the new configuration.



7.	Choose the Start Configuration pushbutton.	 Web Dynpro Explorer: Display Web Dynpro Configuration Konfigurator starten Eigenschaften Configuration Name: Z_FPM_LIST_UIBB Configuration Type: General Config. Variant: Description: Web-Dynpro-Anwendung: Web-Dynpro-Component: FPM_LIST_UIBB Verwaltungsdaten Created By: RUETERL Last Changed By: RUETERL Package: \$TMP Original language: EN
8.	Choose the pushbutton Continue in Change Mode	 Editor for the Web Dynpro ABAP Component Configuration Continue in Change Mode Continue in Display Mode Clear New Other Functions Select the Web Dynpro component that you want to change or display Component Name: FPM_LIST_UIBB Configuration ID: * Z_FPM_LIST_UIBB Advanced Settin... ☐
9.	Under General Settings choose the Feeder Class pushbutton.	 Component Configuration Z_FPM_LIST_UIBB_YMDGM_BUPA Save Cancel Edit Check Additional Functions General Settings Final Flags Feeder Class Feeder Class Parameters Additional Settings

10.	Enter the name of the feeder class.	
11.	To confirm, choose Yes .	
12.	Enter the feeder parameters as shown. Make sure you enter a value for the number of default rows.	

13.	Choose the Column pushbutton.	 Component Configuration Z_FPM_LIST_UIBB_YMDGM_BUPA0 Save Cancel Edit Check Additional Functions General Settings Preview List UIBB Schema Toolbar Schema Menu Schema Lead Selection Action Assignment Column Up Down <table><tr><th>Element</th><th>Display Type</th></tr><tr><td colspan="2">The table does not contain any data</td></tr></table> Quickview Schema	Element	Display Type	The table does not contain any data													
Element	Display Type																	
The table does not contain any data																		
14.	View the list of available columns.	 Edit Columns <table><tr><th colspan="2">Available Columns</th></tr><tr><th>Field Name</th><th>Field Label</th></tr><tr><td>BUPA_ID</td><td>Business Partner</td></tr><tr><td>MATNR</td><td>Material</td></tr><tr><td>MATNR__TXT</td><td>Material Description</td></tr><tr><td>MATNR__TXT02</td><td>Material description</td></tr><tr><td>NICKNAME</td><td>Nickname/name used</td></tr><tr><td>FPM_ROW_ACTIONS_COLU...</td><td>FPM_ROW_ACTIONS_COLU...</td></tr></table> OK Cancel	Available Columns		Field Name	Field Label	BUPA_ID	Business Partner	MATNR	Material	MATNR__TXT	Material Description	MATNR__TXT02	Material description	NICKNAME	Nickname/name used	FPM_ROW_ACTIONS_COLU...	FPM_ROW_ACTIONS_COLU...
Available Columns																		
Field Name	Field Label																	
BUPA_ID	Business Partner																	
MATNR	Material																	
MATNR__TXT	Material Description																	
MATNR__TXT02	Material description																	
NICKNAME	Nickname/name used																	
FPM_ROW_ACTIONS_COLU...	FPM_ROW_ACTIONS_COLU...																	

15. Select the columns from the list of available columns as shown.

Component Configuration Z_FPM_LIST_UIBB_YMDGM_BUPA

Save | Cancel | Edit | Check | Additional Functions

General Settings

Preview

List UIBB Schema | Toolbar Schema | Menu Schema

Lead Selection Action Assignment

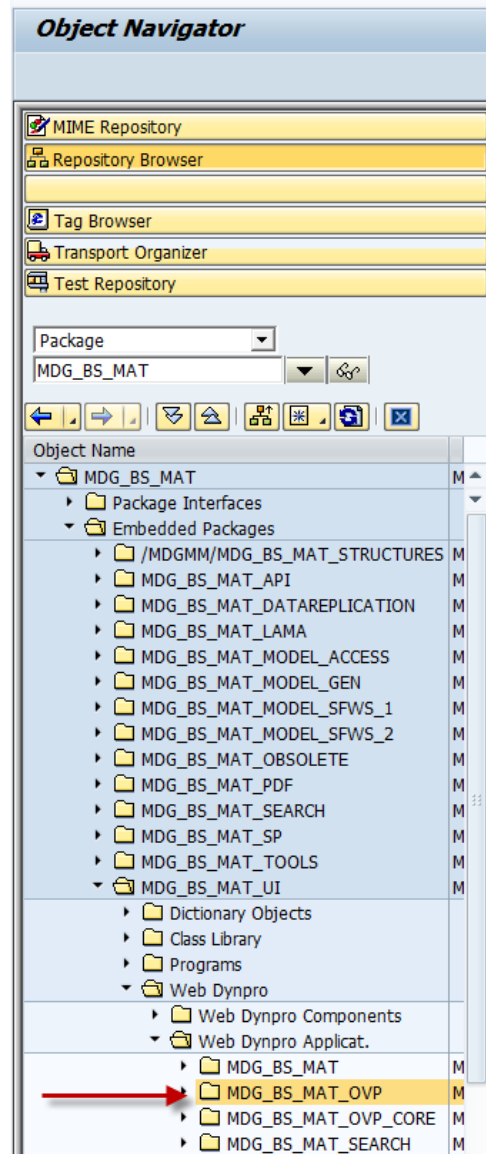
Column | Up | Down

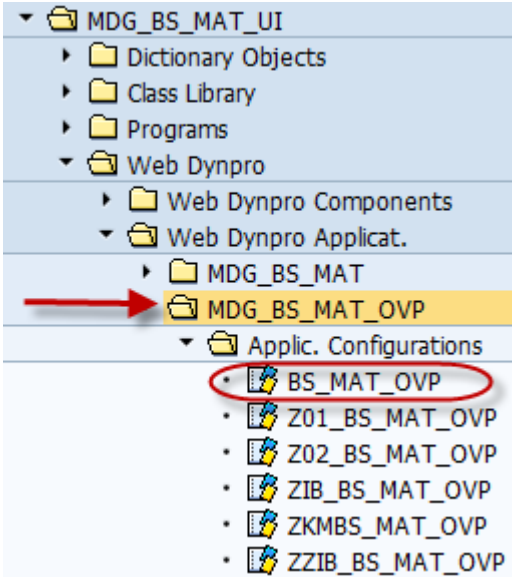
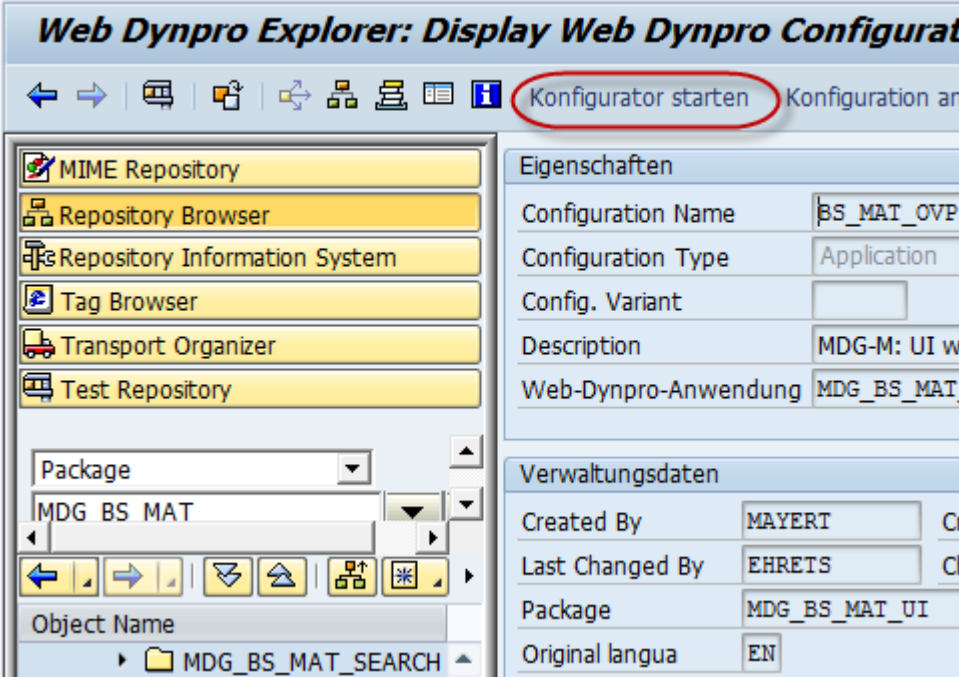
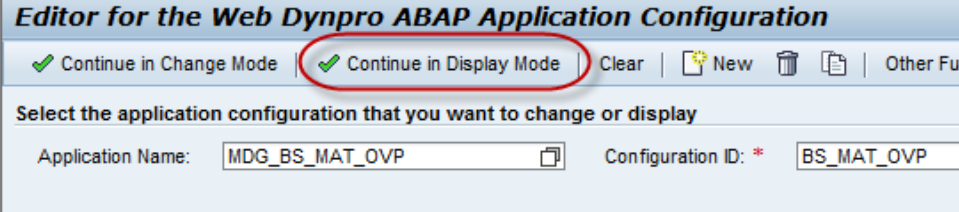
Element	Display Type	Header
Column: MATNR	Input Field	Material
Column: MATNR__TXT	Input Field	Material Description
Column: BUPA_ID	Input Field	Business Partner
Column: NICKNAME	Input Field	Nickname/name us

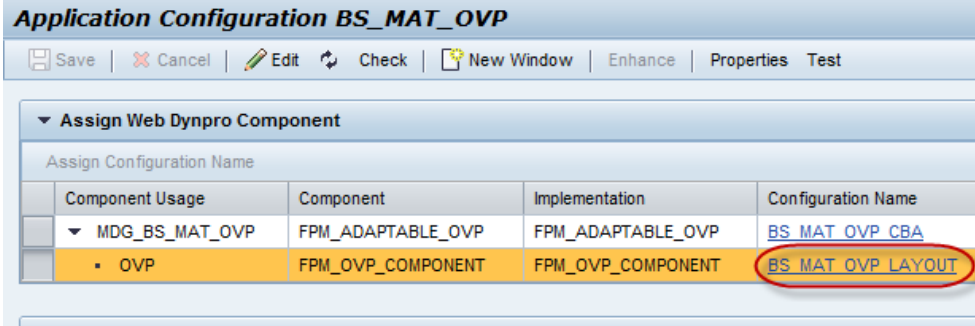
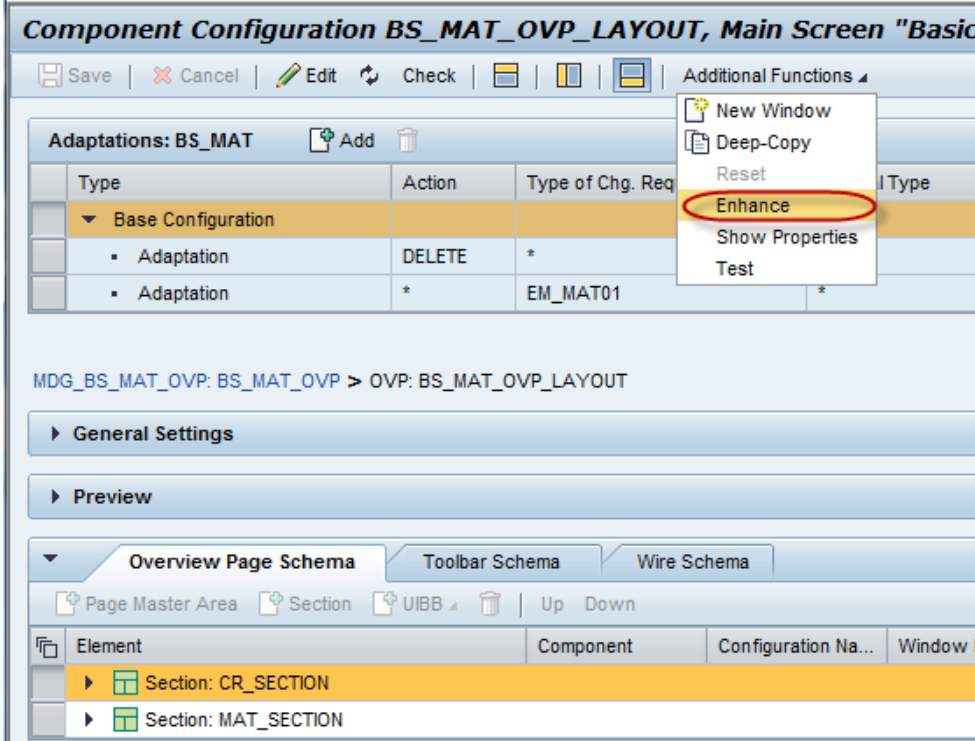
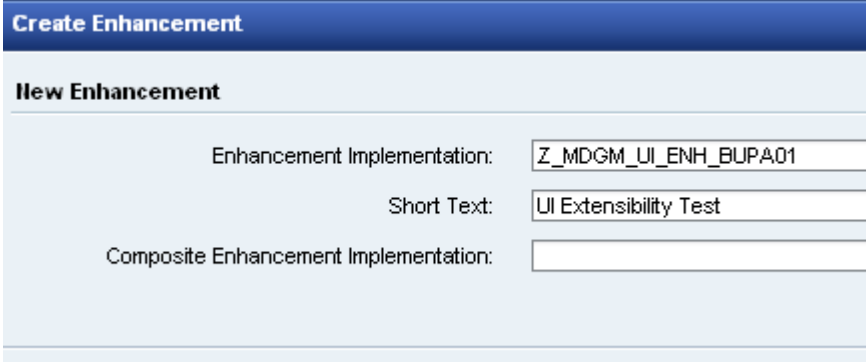
Quickview Schema

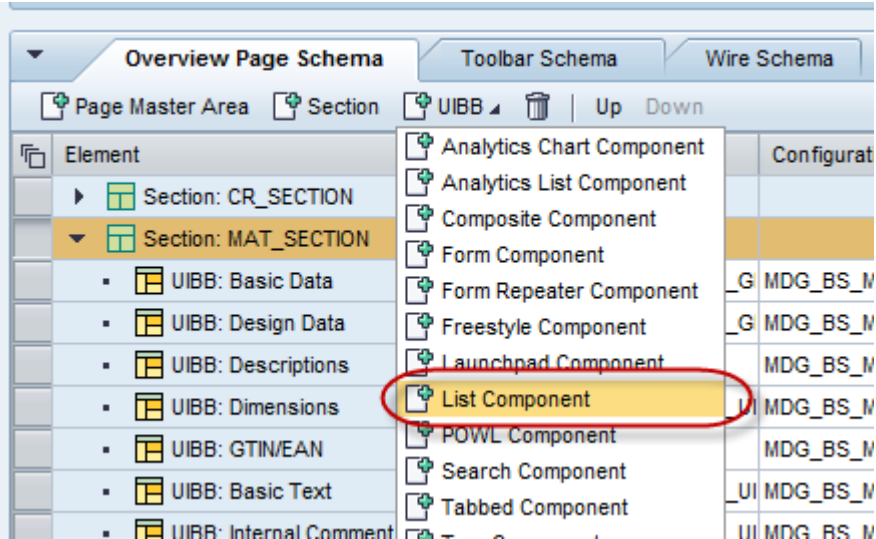
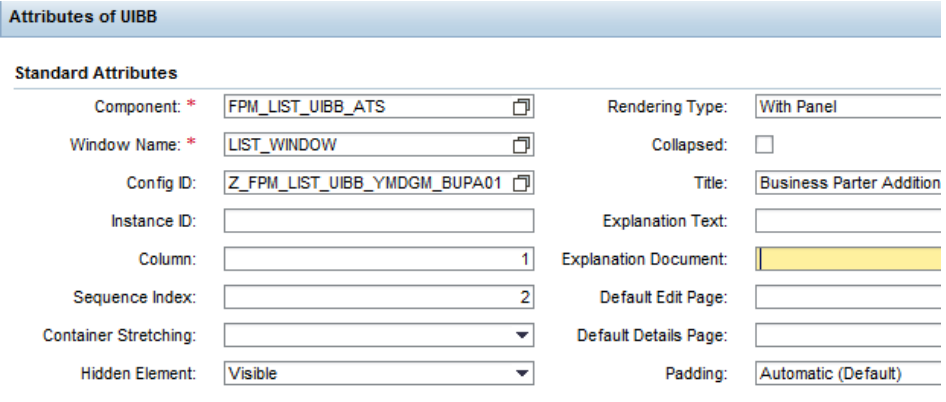
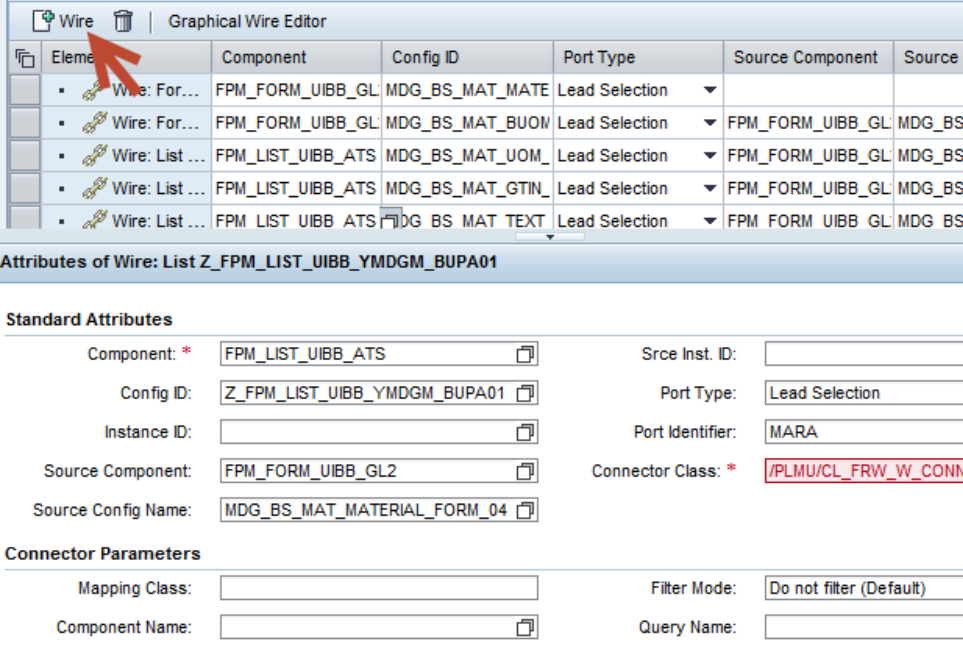
5.5.2. Add List UIBB to Material UI

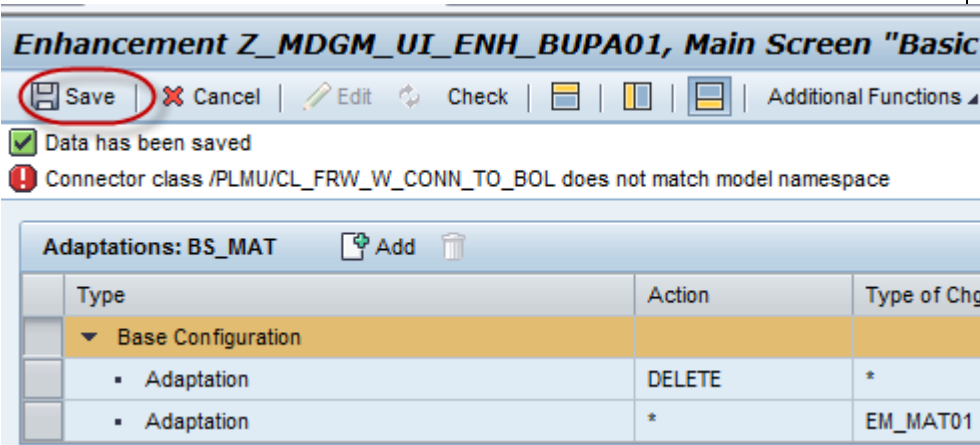
1. Navigate to the **Application Configuration** as shown.



2.	Open the configuration BS_MAT_OVP .	
3.	Choose the Start Configuration pushbutton.	
4.	Choose the Continue in Display Mode pushbutton.	

5.	Navigate to the configuration BS_MAT_OVP_LAYOUT . UT.	 Application Configuration BS_MAT_OVP Save Cancel Edit Check New Window Enhance Properties Test ▼ Assign Web Dynpro Component Assign Configuration Name <table><thead><tr><th></th><th>Component Usage</th><th>Component</th><th>Implementation</th><th>Configuration Name</th></tr></thead><tbody><tr><td></td><td>▼ MDG_BS_MAT_OVP</td><td>FPM_ADAPTABLE_OVP</td><td>FPM_ADAPTABLE_OVP</td><td>BS_MAT_OVP_CBA</td></tr><tr><td></td><td>▪ OVP</td><td>FPM_OVP_COMPONENT</td><td>FPM_OVP_COMPONENT</td><td>BS_MAT_OVP_LAYOUT</td></tr></tbody></table>		Component Usage	Component	Implementation	Configuration Name		▼ MDG_BS_MAT_OVP	FPM_ADAPTABLE_OVP	FPM_ADAPTABLE_OVP	BS_MAT_OVP_CBA		▪ OVP	FPM_OVP_COMPONENT	FPM_OVP_COMPONENT	BS_MAT_OVP_LAYOUT													
	Component Usage	Component	Implementation	Configuration Name																										
	▼ MDG_BS_MAT_OVP	FPM_ADAPTABLE_OVP	FPM_ADAPTABLE_OVP	BS_MAT_OVP_CBA																										
	▪ OVP	FPM_OVP_COMPONENT	FPM_OVP_COMPONENT	BS_MAT_OVP_LAYOUT																										
6.	Select Additional Functions -> Enhance . Or you can create a new CBA or you can use customizing mode.	 Component Configuration BS_MAT_OVP_LAYOUT, Main Screen "Basic" Save Cancel Edit Check Additional Functions Adaptations: BS_MAT <table><thead><tr><th>Type</th><th>Action</th><th>Type of Chg. Req.</th><th>Type</th></tr></thead><tbody><tr><td>▼ Base Configuration</td><td></td><td></td><td></td></tr><tr><td>▪ Adaptation</td><td>DELETE</td><td>*</td><td></td></tr><tr><td>▪ Adaptation</td><td>*</td><td>EM_MAT01</td><td>*</td></tr></tbody></table> MDG_BS_MAT_OVP: BS_MAT_OVP > OVP: BS_MAT_OVP_LAYOUT ► General Settings ► Preview ▼ Overview Page Schema Toolbar Schema Wire Schema Page Master Area Section UIBB Up Down <table><thead><tr><th>Element</th><th>Component</th><th>Configuration Na...</th><th>Window M</th></tr></thead><tbody><tr><td>► Section: CR_SECTION</td><td></td><td></td><td></td></tr><tr><td>► Section: MAT_SECTION</td><td></td><td></td><td></td></tr></tbody></table>	Type	Action	Type of Chg. Req.	Type	▼ Base Configuration				▪ Adaptation	DELETE	*		▪ Adaptation	*	EM_MAT01	*	Element	Component	Configuration Na...	Window M	► Section: CR_SECTION				► Section: MAT_SECTION			
Type	Action	Type of Chg. Req.	Type																											
▼ Base Configuration																														
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▪ Adaptation	*	EM_MAT01	*																											
Element	Component	Configuration Na...	Window M																											
► Section: CR_SECTION																														
► Section: MAT_SECTION																														
7.	Provide a name for your enhancement.	 Create Enhancement New Enhancement Enhancement Implementation: Z_MDGM_UI_ENH_BUPA01 Short Text: UI Extensibility Test Composite Enhancement Implementation:																												

8.	Add a new List Component to the Section : MAT_SECTION element as shown.	 The screenshot shows the 'Overview Page Schema' with a tree view on the left and a component list on the right. The tree view shows 'Section: MAT_SECTION' expanded, with sub-elements like 'UIBB: Basic Data', 'UIBB: Design Data', etc. The right pane shows a list of components, with 'List Component' highlighted in yellow and circled in red.																																																																		
9.	Enter the details for the new list component.	 The screenshot shows the 'Attributes of UIBB' dialog box. The 'Standard Attributes' section contains the following fields: <table><tr><th>Attribute</th><th>Value</th></tr><tr><td>Component: *</td><td>FPM_LIST_UIBB_ATS</td></tr><tr><td>Window Name: *</td><td>LIST_WINDOW</td></tr><tr><td>Config ID:</td><td>Z_FPM_LIST_UIBB_YMDGM_BUPA01</td></tr><tr><td>Instance ID:</td><td></td></tr><tr><td>Column:</td><td>1</td></tr><tr><td>Sequence Index:</td><td>2</td></tr><tr><td>Container Stretching:</td><td></td></tr><tr><td>Hidden Element:</td><td>Visible</td></tr><tr><td>Rendering Type:</td><td>With Panel</td></tr><tr><td>Collapsed:</td><td>☐</td></tr><tr><td>Title:</td><td>Business Partner Addition</td></tr><tr><td>Explanation Text:</td><td></td></tr><tr><td>Explanation Document:</td><td></td></tr><tr><td>Default Edit Page:</td><td></td></tr><tr><td>Default Details Page:</td><td></td></tr><tr><td>Padding:</td><td>Automatic (Default)</td></tr></table>	Attribute	Value	Component: *	FPM_LIST_UIBB_ATS	Window Name: *	LIST_WINDOW	Config ID:	Z_FPM_LIST_UIBB_YMDGM_BUPA01	Instance ID:		Column:	1	Sequence Index:	2	Container Stretching:		Hidden Element:	Visible	Rendering Type:	With Panel	Collapsed:	☐	Title:	Business Partner Addition	Explanation Text:		Explanation Document:		Default Edit Page:		Default Details Page:		Padding:	Automatic (Default)																																
Attribute	Value																																																																			
Component: *	FPM_LIST_UIBB_ATS																																																																			
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Instance ID:																																																																				
Column:	1																																																																			
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Explanation Document:																																																																				
Default Edit Page:																																																																				
Default Details Page:																																																																				
Padding:	Automatic (Default)																																																																			
10.	Create a new wire entry with the details shown in the screenshot. Use connector class PLMU/CL_FRW_W_CONN_DEFAULT	 The screenshot shows the 'Wire' dialog box with a table of existing wires and the 'Attributes of Wire' section. <table><tr><th>Element</th><th>Component</th><th>Config ID</th><th>Port Type</th><th>Source Component</th><th>Source Config</th></tr><tr><td>Wire: For...</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_MAT_MATE</td><td>Lead Selection</td><td></td><td></td></tr><tr><td>Wire: For...</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_MAT_BUOM</td><td>Lead Selection</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_I</td></tr><tr><td>Wire: List ...</td><td>FPM_LIST_UIBB_ATS</td><td>MDG_BS_MAT_UOM</td><td>Lead Selection</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_I</td></tr><tr><td>Wire: List ...</td><td>FPM_LIST_UIBB_ATS</td><td>MDG_BS_MAT_GTIN</td><td>Lead Selection</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_I</td></tr><tr><td>Wire: List ...</td><td>FPM_LIST_UIBB_ATS</td><td>MDG_BS_MAT_TEXT</td><td>Lead Selection</td><td>FPM_FORM_UIBB_GL</td><td>MDG_BS_I</td></tr></table> The 'Attributes of Wire' section contains the following fields: <table><tr><th>Attribute</th><th>Value</th></tr><tr><td>Component: *</td><td>FPM_LIST_UIBB_ATS</td></tr><tr><td>Config ID:</td><td>Z_FPM_LIST_UIBB_YMDGM_BUPA01</td></tr><tr><td>Instance ID:</td><td></td></tr><tr><td>Source Component:</td><td>FPM_FORM_UIBB_GL2</td></tr><tr><td>Source Config Name:</td><td>MDG_BS_MAT_MATERIAL_FORM_04</td></tr><tr><td>Src Inst. ID:</td><td></td></tr><tr><td>Port Type:</td><td>Lead Selection</td></tr><tr><td>Port Identifier:</td><td>MARA</td></tr><tr><td>Connector Class: *</td><td>/PLMU/CL_FRW_W_CONN_</td></tr></table> The 'Connector Parameters' section contains the following fields: <table><tr><th>Attribute</th><th>Value</th></tr><tr><td>Mapping Class:</td><td></td></tr><tr><td>Component Name:</td><td></td></tr><tr><td>Filter Mode:</td><td>Do not filter (Default)</td></tr><tr><td>Query Name:</td><td></td></tr></table>	Element	Component	Config ID	Port Type	Source Component	Source Config	Wire: For...	FPM_FORM_UIBB_GL	MDG_BS_MAT_MATE	Lead Selection			Wire: For...	FPM_FORM_UIBB_GL	MDG_BS_MAT_BUOM	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I	Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_UOM	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I	Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_GTIN	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I	Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_TEXT	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I	Attribute	Value	Component: *	FPM_LIST_UIBB_ATS	Config ID:	Z_FPM_LIST_UIBB_YMDGM_BUPA01	Instance ID:		Source Component:	FPM_FORM_UIBB_GL2	Source Config Name:	MDG_BS_MAT_MATERIAL_FORM_04	Src Inst. ID:		Port Type:	Lead Selection	Port Identifier:	MARA	Connector Class: *	/PLMU/CL_FRW_W_CONN_	Attribute	Value	Mapping Class:		Component Name:		Filter Mode:	Do not filter (Default)	Query Name:	
Element	Component	Config ID	Port Type	Source Component	Source Config																																																															
Wire: For...	FPM_FORM_UIBB_GL	MDG_BS_MAT_MATE	Lead Selection																																																																	
Wire: For...	FPM_FORM_UIBB_GL	MDG_BS_MAT_BUOM	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I																																																															
Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_UOM	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I																																																															
Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_GTIN	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I																																																															
Wire: List ...	FPM_LIST_UIBB_ATS	MDG_BS_MAT_TEXT	Lead Selection	FPM_FORM_UIBB_GL	MDG_BS_I																																																															
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Source Component:	FPM_FORM_UIBB_GL2																																																																			
Source Config Name:	MDG_BS_MAT_MATERIAL_FORM_04																																																																			
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Attribute	Value																																																																			
Mapping Class:																																																																				
Component Name:																																																																				
Filter Mode:	Do not filter (Default)																																																																			
Query Name:																																																																				

11.	Save your changes.	 The screenshot shows the SAP Enhancement Z_MDGM_UI_ENH_BUPA01, Main Screen 'Basic'. The 'Save' button is highlighted with a red circle. Below the toolbar, a message states 'Data has been saved'. Below that, a warning message indicates 'Connector class /PLMU/CL_FRW_W_CONN_TO_BOL does not match model namespace'. At the bottom, there is a table titled 'Adaptations: BS_MAT' with columns 'Type', 'Action', and 'Type of Change'. <table border="1"> <thead> <tr> <th>Type</th> <th>Action</th> <th>Type of Change</th> </tr> </thead> <tbody> <tr> <td>Base Configuration</td> <td></td> <td></td> </tr> <tr> <td>Adaptation</td> <td>DELETE</td> <td>*</td> </tr> <tr> <td>Adaptation</td> <td>*</td> <td>EM_MAT01</td> </tr> </tbody> </table>	Type	Action	Type of Change	Base Configuration			Adaptation	DELETE	*	Adaptation	*	EM_MAT01
Type	Action	Type of Change												
Base Configuration														
Adaptation	DELETE	*												
Adaptation	*	EM_MAT01												

5.5.3. Clear UI Metadata Buffers

After finishing the UI, clear the metadata buffers. You can find the report in customizing Master Data Governance, Central Governance-> Master Data Governance for Material-> Clear UI Metadata Buffers.

This report clears the following buffers in this sequence:

- Text Buffer
- Search Help Buffer
- SMT-Mapping Data Buffer
- SPI Metadata Buffer

You should use this report after extending the data model to make sure that the metadata is consistent with the MDG customizing and UI configuration.

6. Testing Your Data Model Extension

To test your configuration, start the MDG Material UI using the following URL (replace the parameters host, port and client-id to match your landscape):

`https://<host>:<port>/sap/bc/webdynpro/sap/mdg_bs_mat?ACTION=CREATE&WDCONFIGURATIONID=BS_MAT_INIT&sap-client=<client-id>`

Alternatively, start transaction PFCG, enter role name **SAP_MDGM_MENU** and click the **Display** pushbutton. Select the **Menu** tab. In the hierarchy window, navigate to **Role Menu -> Material Governance -> Material Processing**. Right click on **Create Material** and select **Execute** from the drop-down.

Material: SDFGFDG234 Side Panel

Save | Cancel | Edit [Icons]

▼ Change Request

General | Notes | Attachments

General Data	Process Data
Change Request ID: 0	Status:
Description:	Current Workitem:
Priority:	Created On/By: *00.00.0000 00:00:0
Due Date:	
Reason:	
CR Type:	
Type of Change Requ...	

► Basic Data

► Design Data

► Descriptions

► Dimensions

► GTIN/EAN

► Basic Text

► Internal Comment

► Classification

▼ Business Partner Additional Data

Material	Material Description	Business Partner	Nickname/name used
SDFGFDG234		153	TEST2
SDFGFDG234		240	TEST1



7. Related Content

- For more information, visit the Configuration and Enhancement of SAP Master Data Governance homepage. (<http://scn.sap.com/docs/DOC-785>)

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